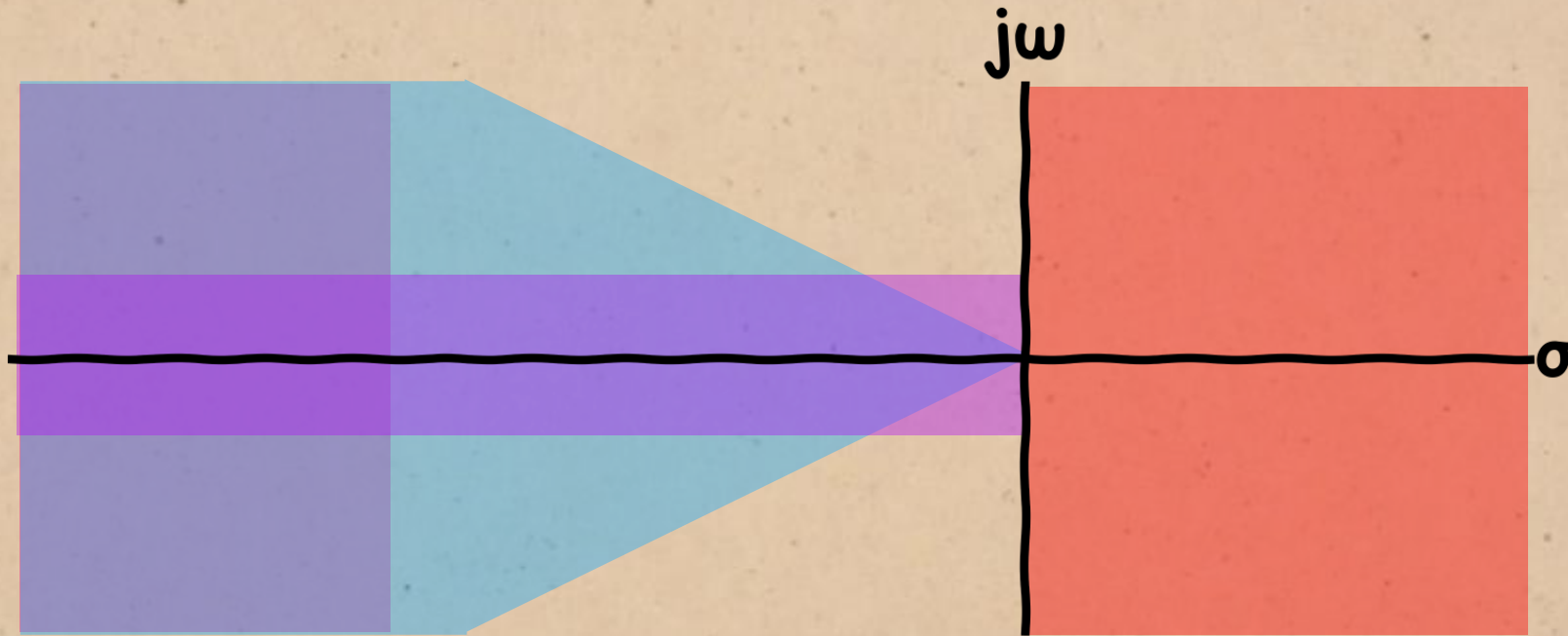


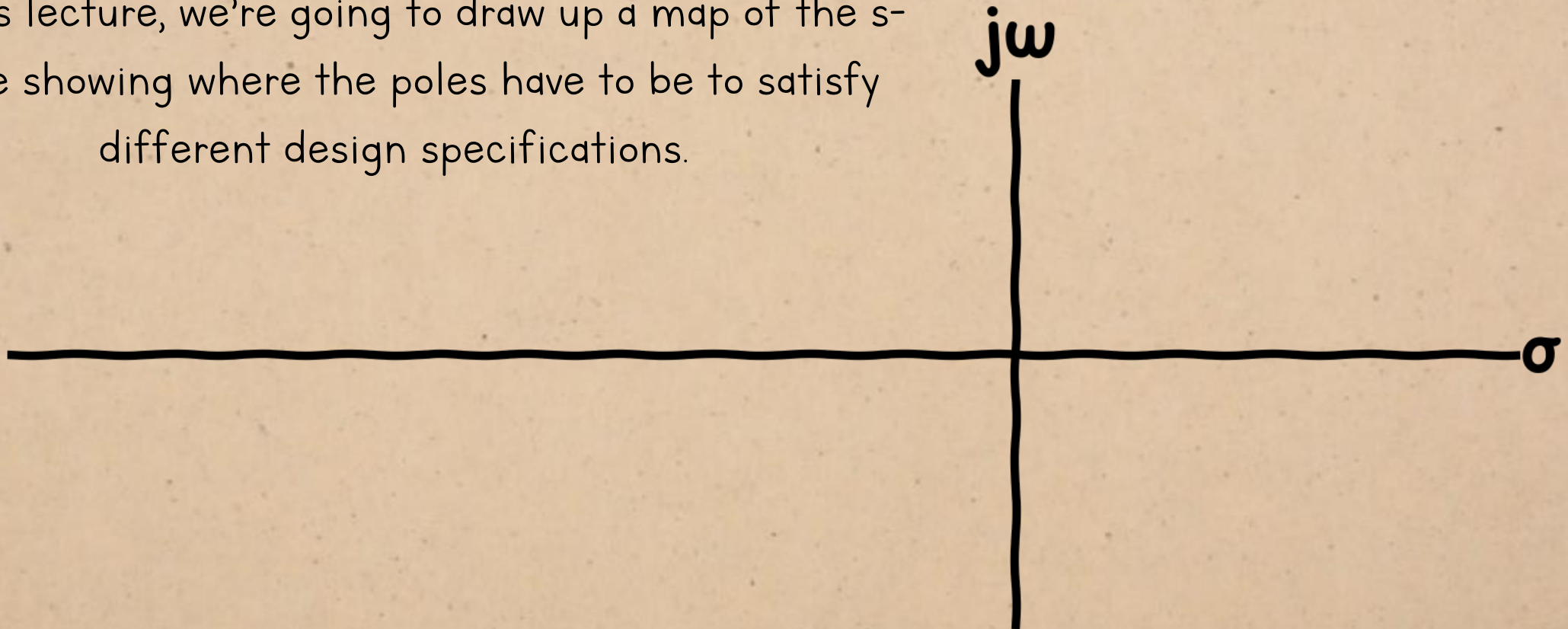


CHAPTER 3: S-PLANE DESIGN

DESIGN GUIDE TO THE S-PLANE

DESIGN GUIDE TO THE S-PLANE

In this lecture, we're going to draw up a map of the s-plane showing where the poles have to be to satisfy different design specifications.



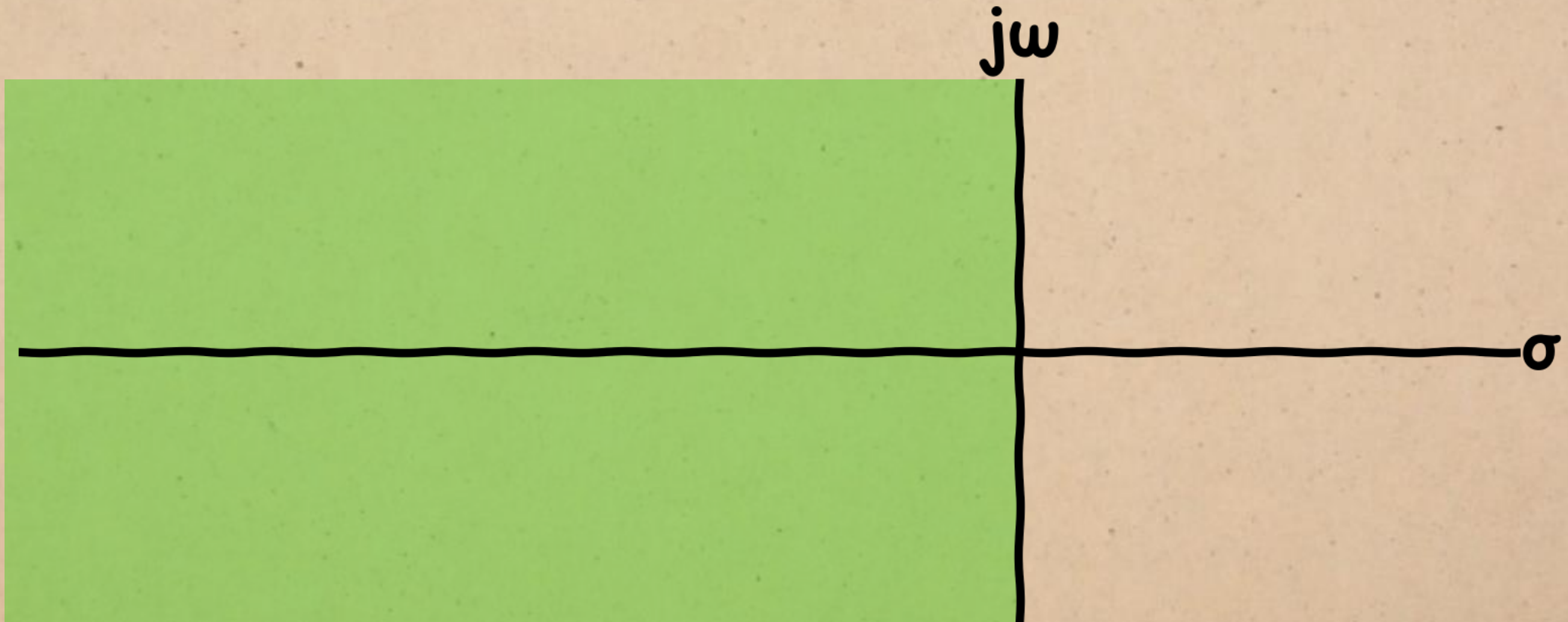
DESIGN GUIDE TO THE S-PLANE

Obviously, the whole right-half plane is forbidden.



where unstable poles lie

DESIGN GUIDE TO THE S-PLANE



...so the poles have to fall in the left-half plane.
But we also care about how far to the left they are.

settling time specifications:

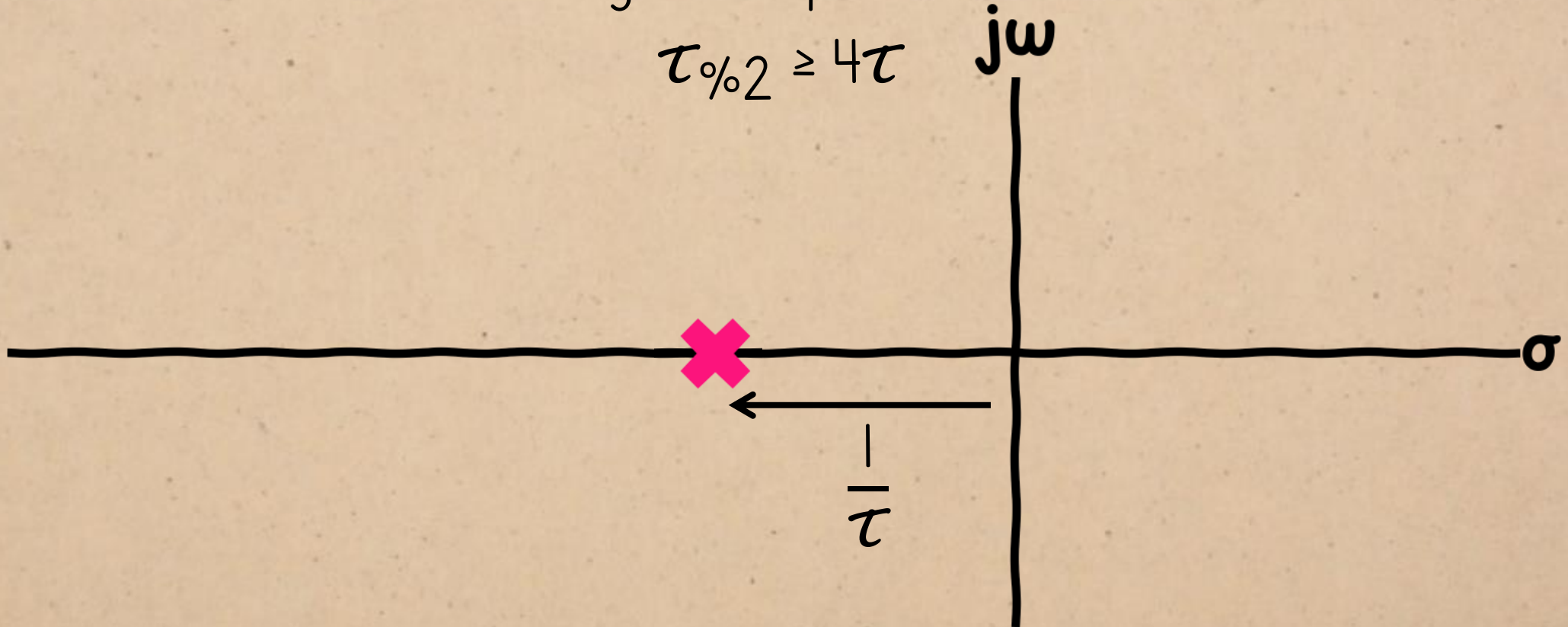


“I need the doohicky to
do whatever it does in
no more than $\tau/2$
seconds!”

DESIGN GUIDE TO THE S-PLANE

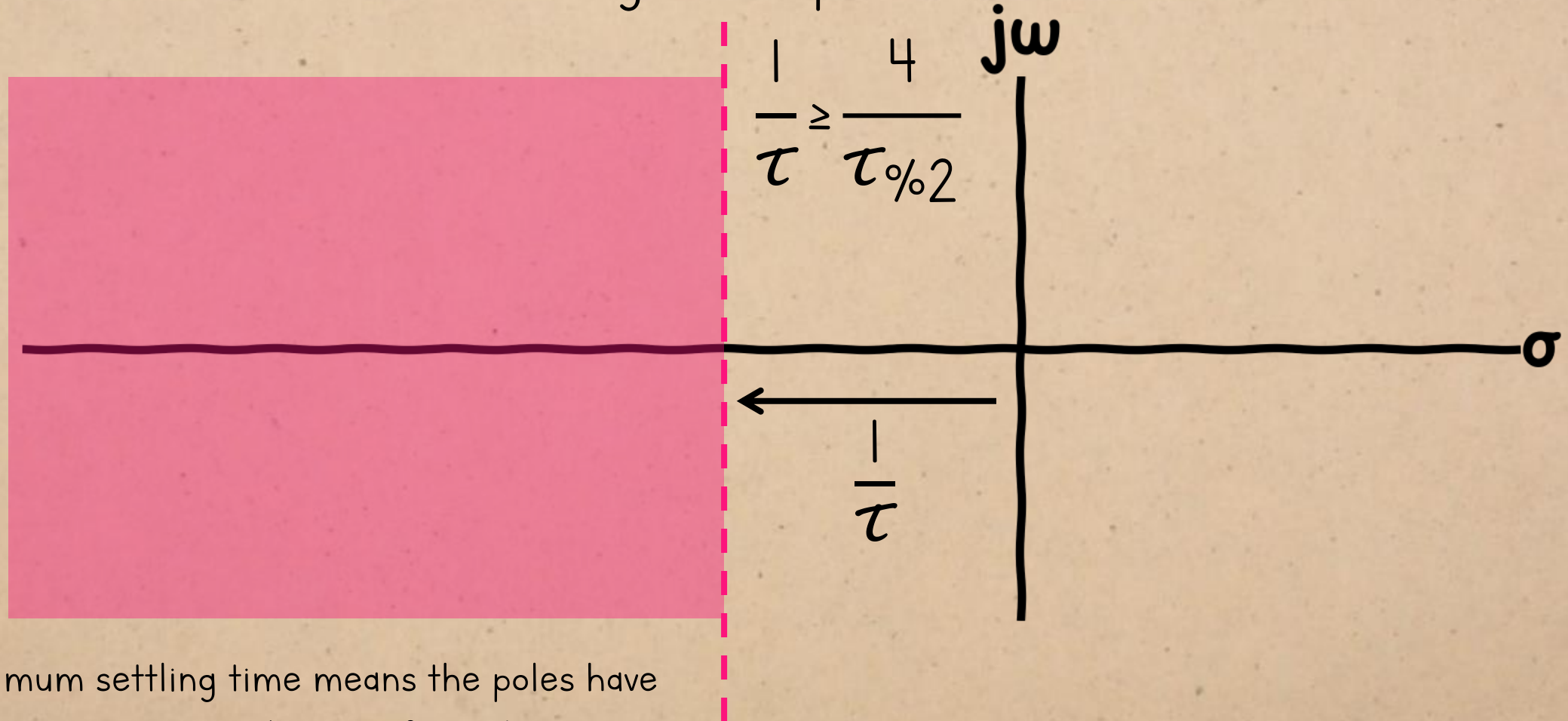
settling time specifications:

$$\tau_{\%2} \geq 4\tau$$



DESIGN GUIDE TO THE S-PLANE

settling time specifications:



A maximum settling time means the poles have to be some minimum distance from the origin.

damping specifications:

“And our focus groups say
there’s **too much oscillation!**”

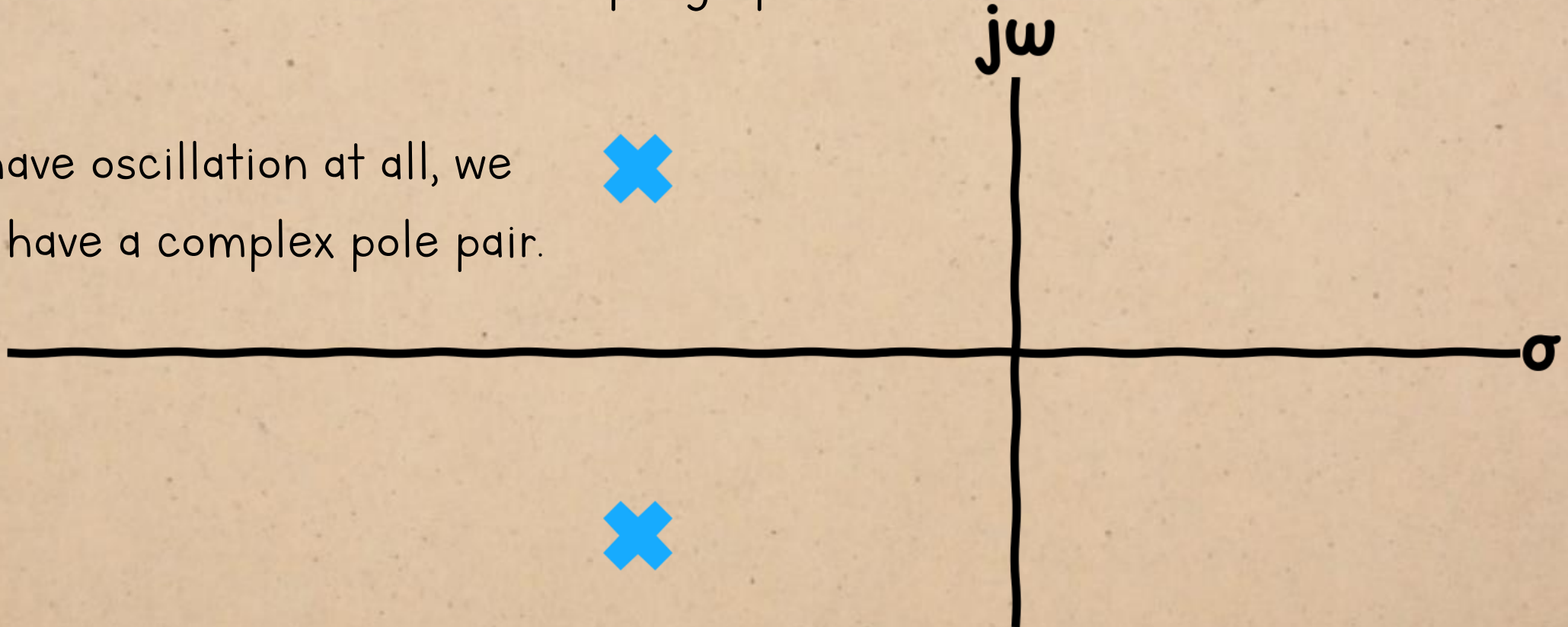
We need
a **damping ratio of $\leq \zeta$**
to get the 18-35 year-old
suburban demographic!”



DESIGN GUIDE TO THE S-PLANE

damping specifications:

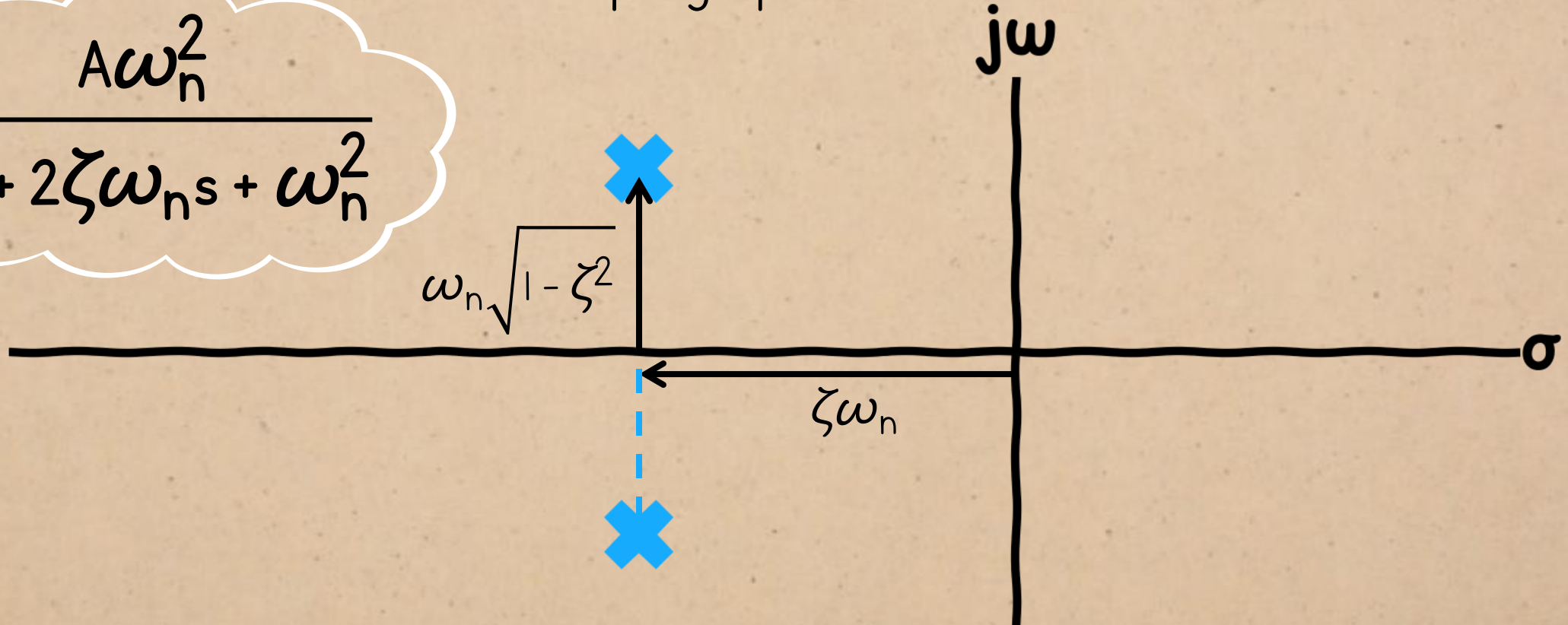
To have oscillation at all, we must have a complex pole pair.



DESIGN GUIDE TO THE S-PLANE

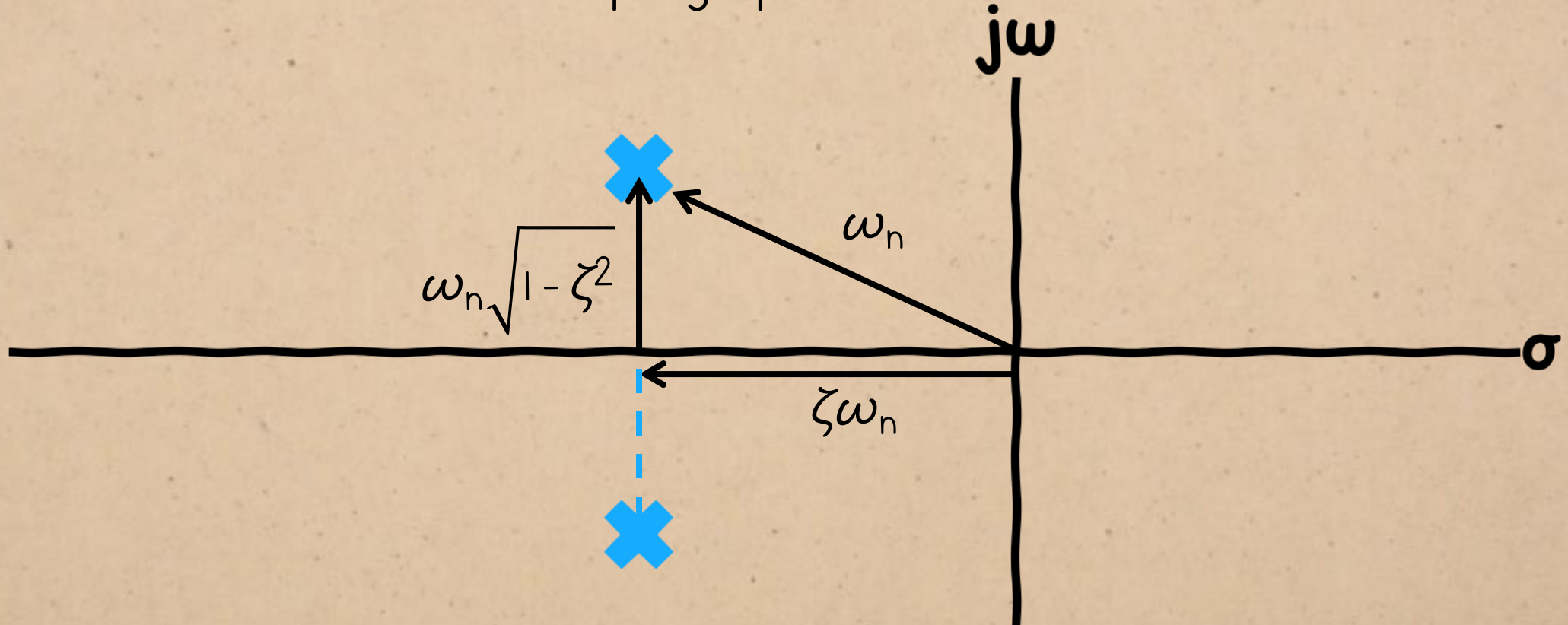
damping specifications:

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



DESIGN GUIDE TO THE S-PLANE

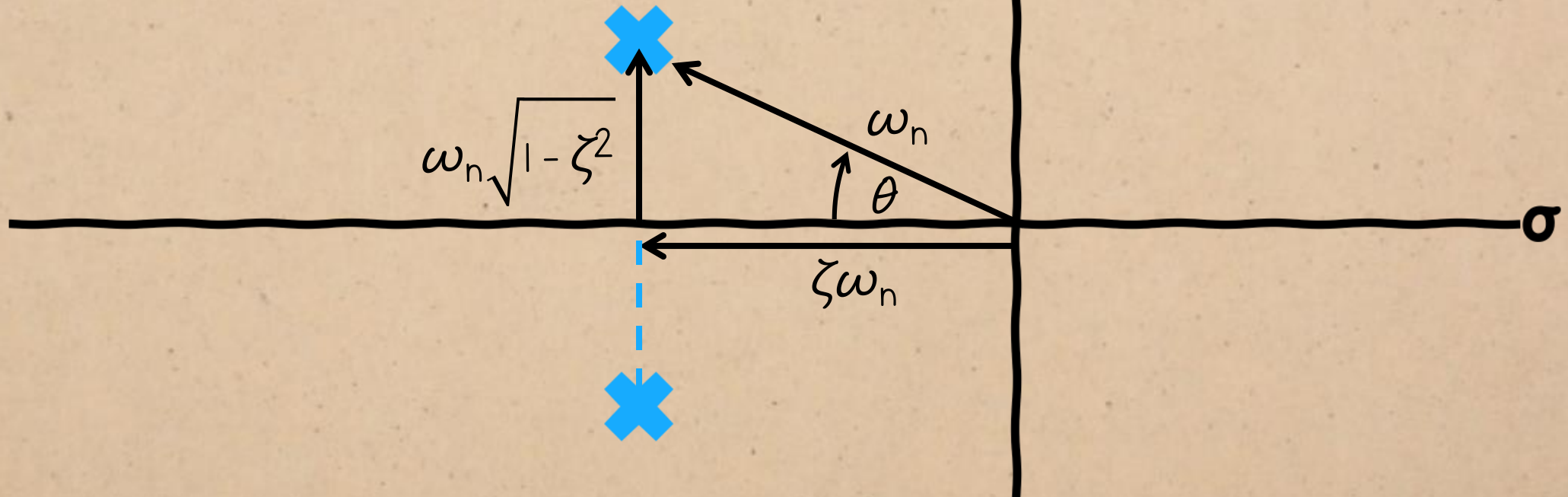
damping specifications:



DESIGN GUIDE TO THE S-PLANE

damping specifications:

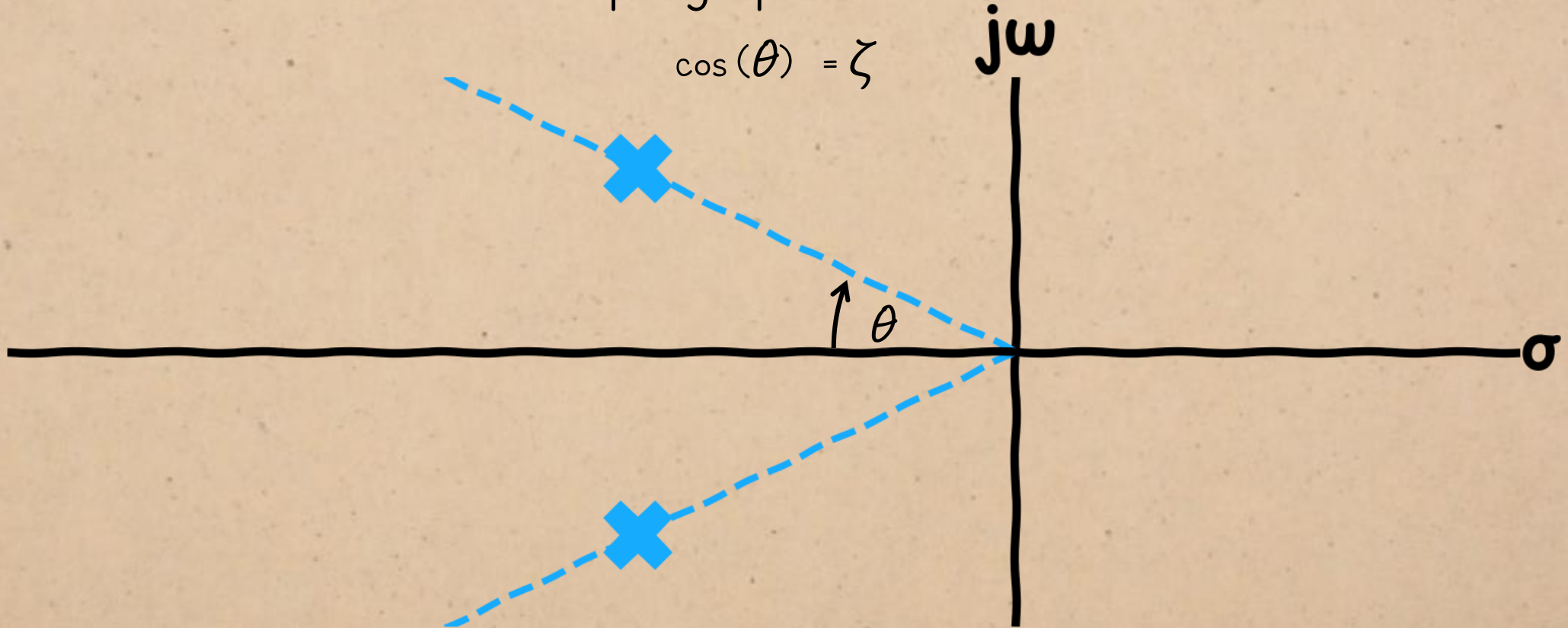
$$\cos(\theta) = \frac{\zeta\omega_n}{\omega_n} \quad j\omega$$



DESIGN GUIDE TO THE S-PLANE

damping specifications:

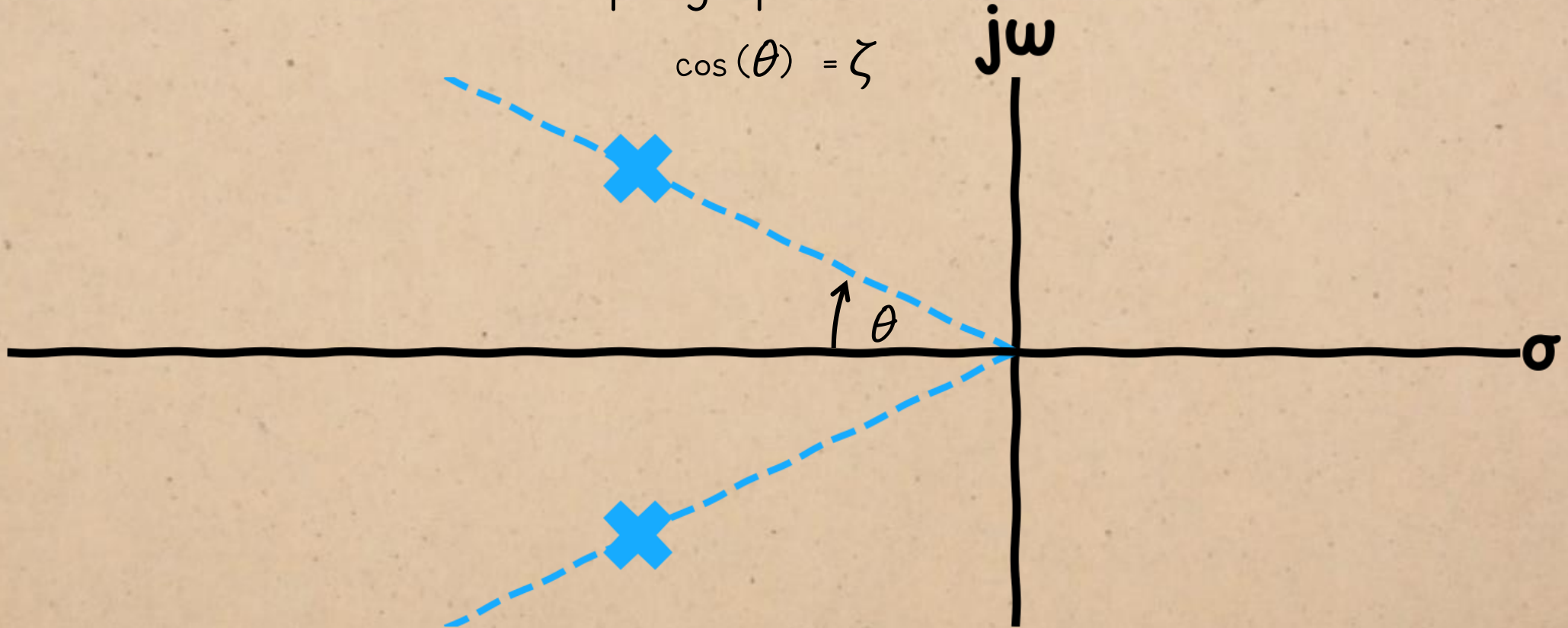
$$\cos(\theta) = \zeta$$



DESIGN GUIDE TO THE S-PLANE

damping specifications:

$$\cos(\theta) = \zeta$$

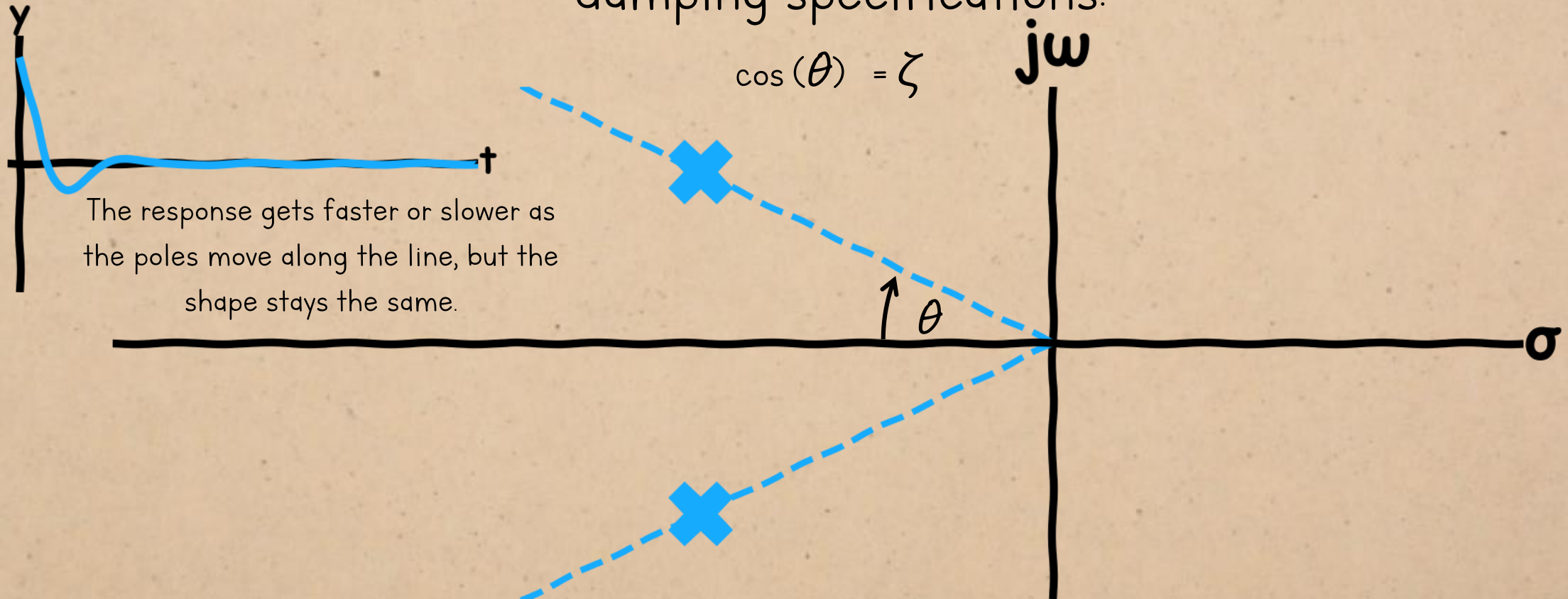


$\therefore$ poles along the same diagonal line will
have the same damping ratio

DESIGN GUIDE TO THE S-PLANE

damping specifications:

$$\cos(\theta) = \zeta$$



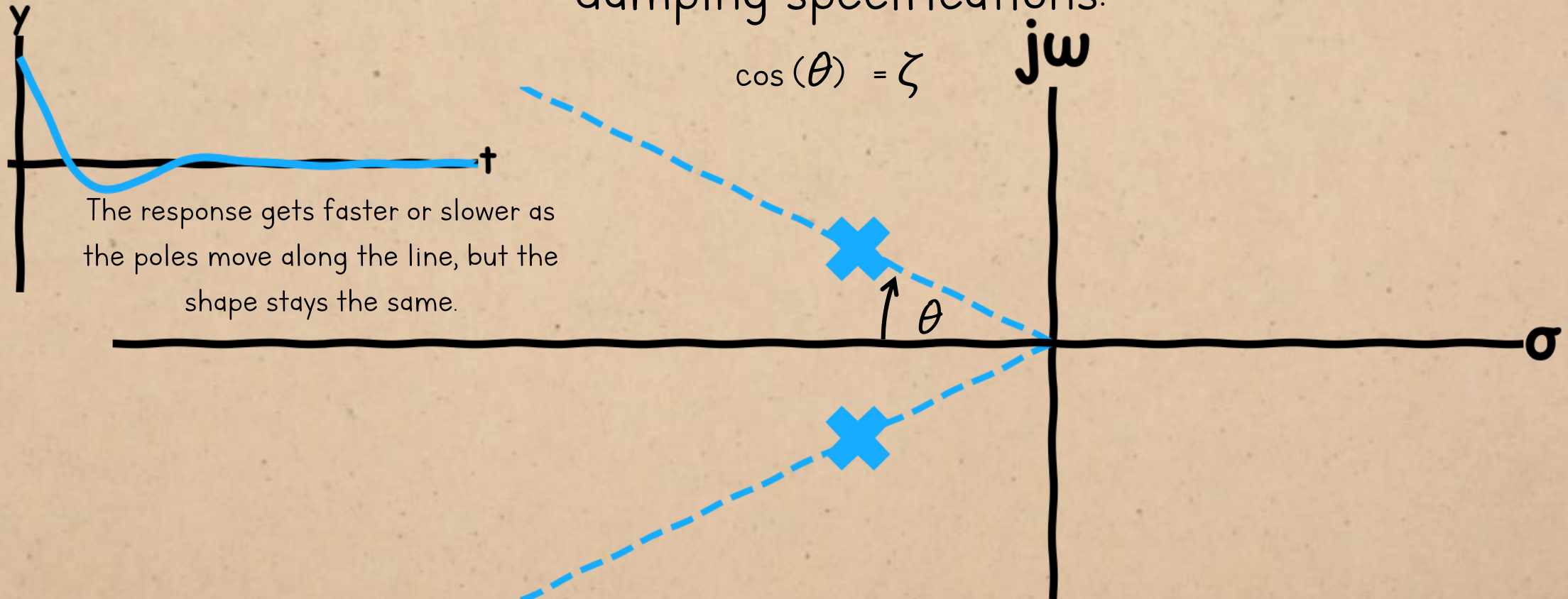
The response gets faster or slower as the poles move along the line, but the shape stays the same.

$\therefore$ poles along the same diagonal line will have the same damping ratio

DESIGN GUIDE TO THE S-PLANE

damping specifications:

$$\cos(\theta) = \zeta$$

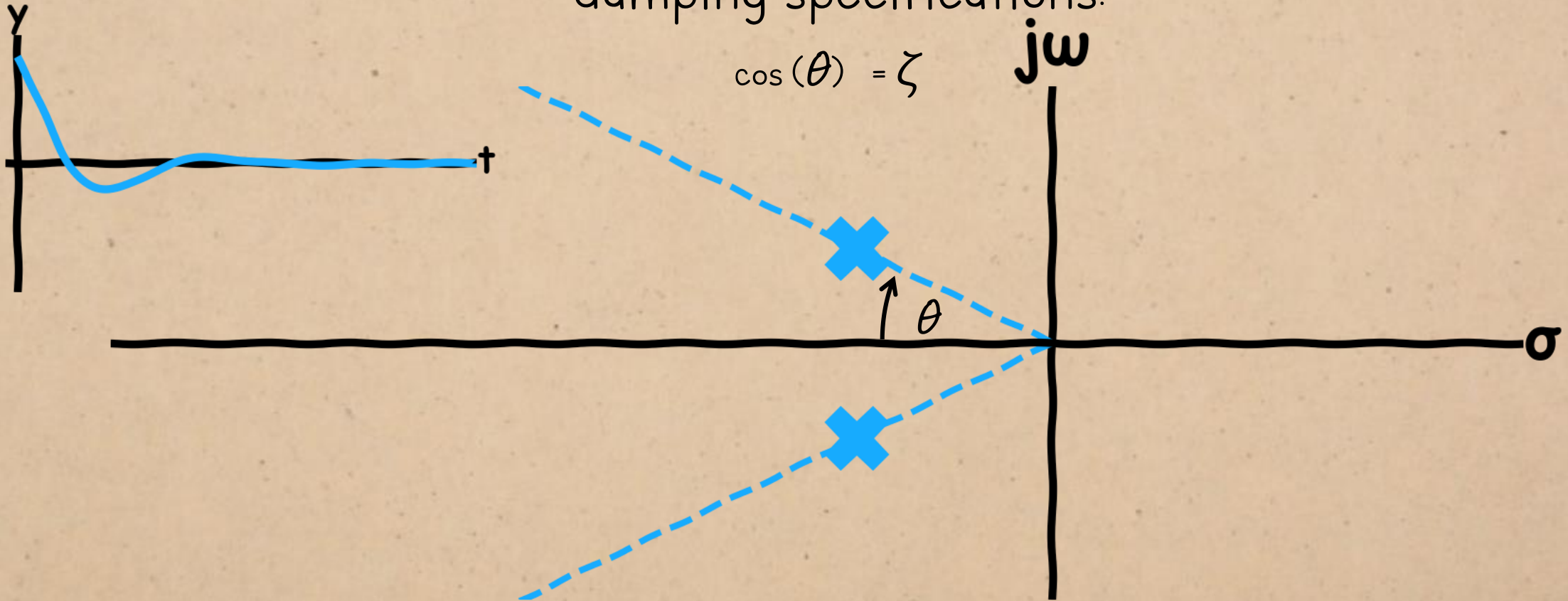


∴ poles along the same diagonal line will
have the same damping ratio

DESIGN GUIDE TO THE S-PLANE

damping specifications:

$$\cos(\theta) = \zeta$$



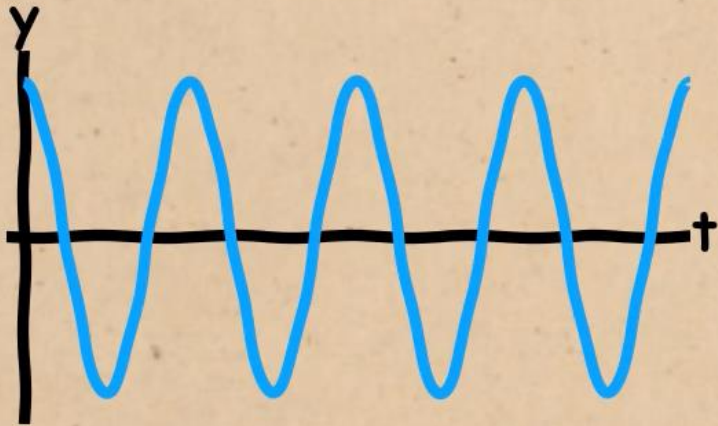
The steeper the angle of the pole is, the more oscillatory the response will be.

DESIGN GUIDE TO THE S-PLANE

damping specifications:

$$\cos(\theta) = \zeta$$

$j\omega$



The steeper the angle of the pole is, the more oscillatory the response will be.

DESIGN GUIDE TO THE S-PLANE

damping specifications:

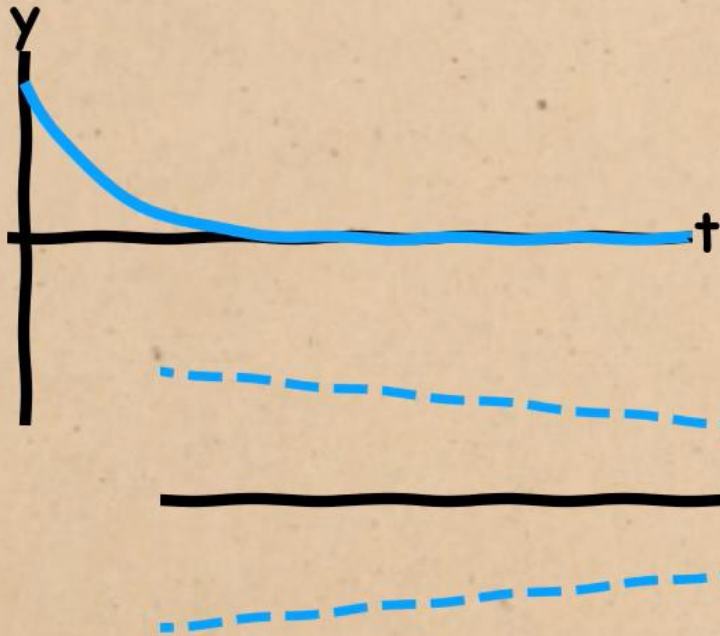
$$\cos(\theta) = \zeta$$

$j\omega$

For many applications, you don't need to avoid complex poles altogether. A response like this, which is underdamped but not oscillatory, is often acceptable.

Rule of thumb: I'd recommend using a damping ratio of around 0.7 as a lower limit. This corresponds to a pole angle of 45° , so the real and imaginary parts are equal.

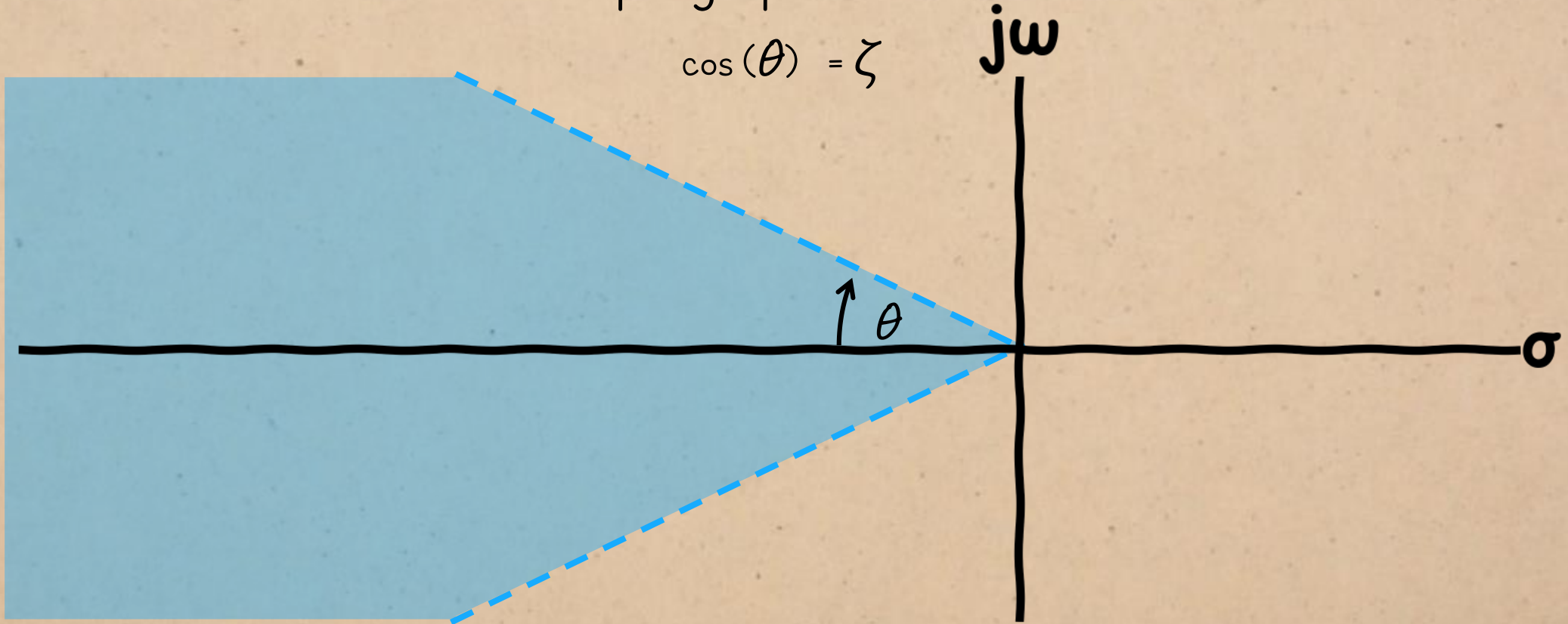
When the poles are close to the real axis, the response looks more like the critically damped response.



DESIGN GUIDE TO THE S-PLANE

damping specifications:

$$\cos(\theta) = \zeta$$



A damping specification is represented by a wedge-shaped region.

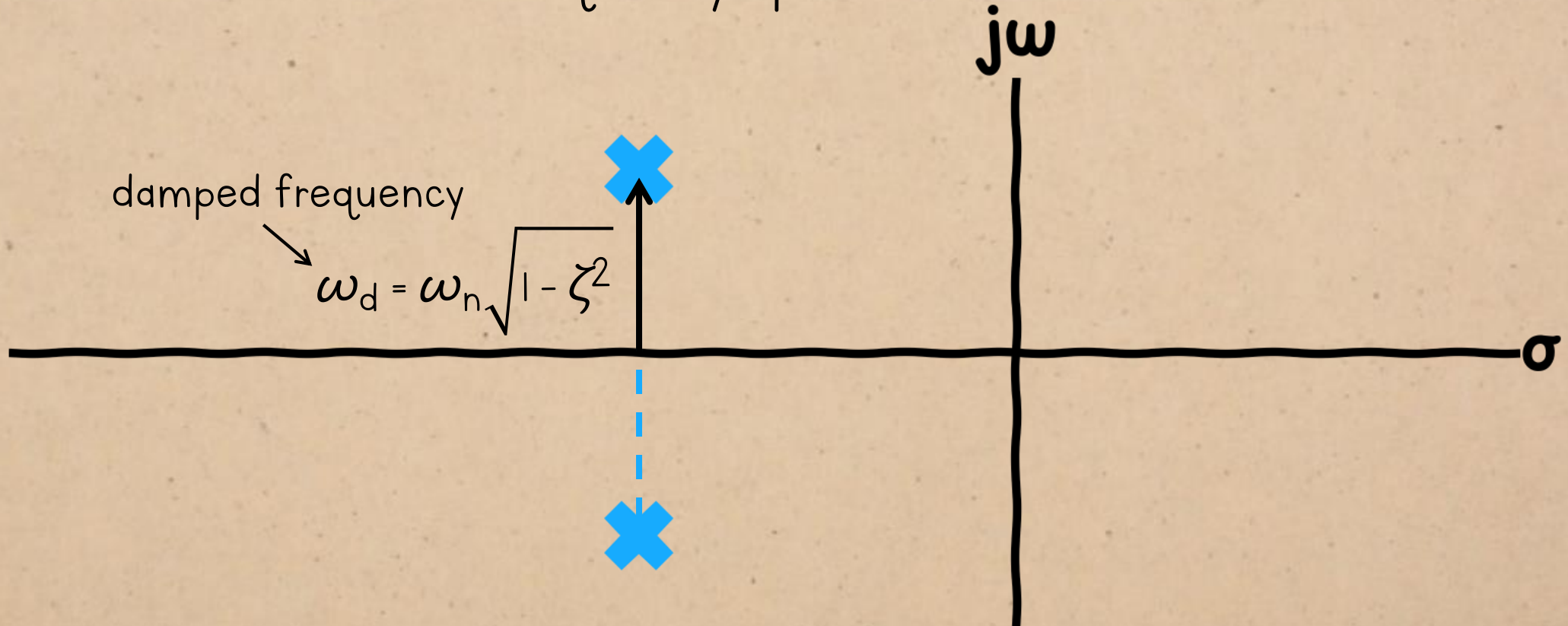
frequency specifications:



“And the Tokyo deal is
dead in the water unless
we keep the
frequency below ω_d !”

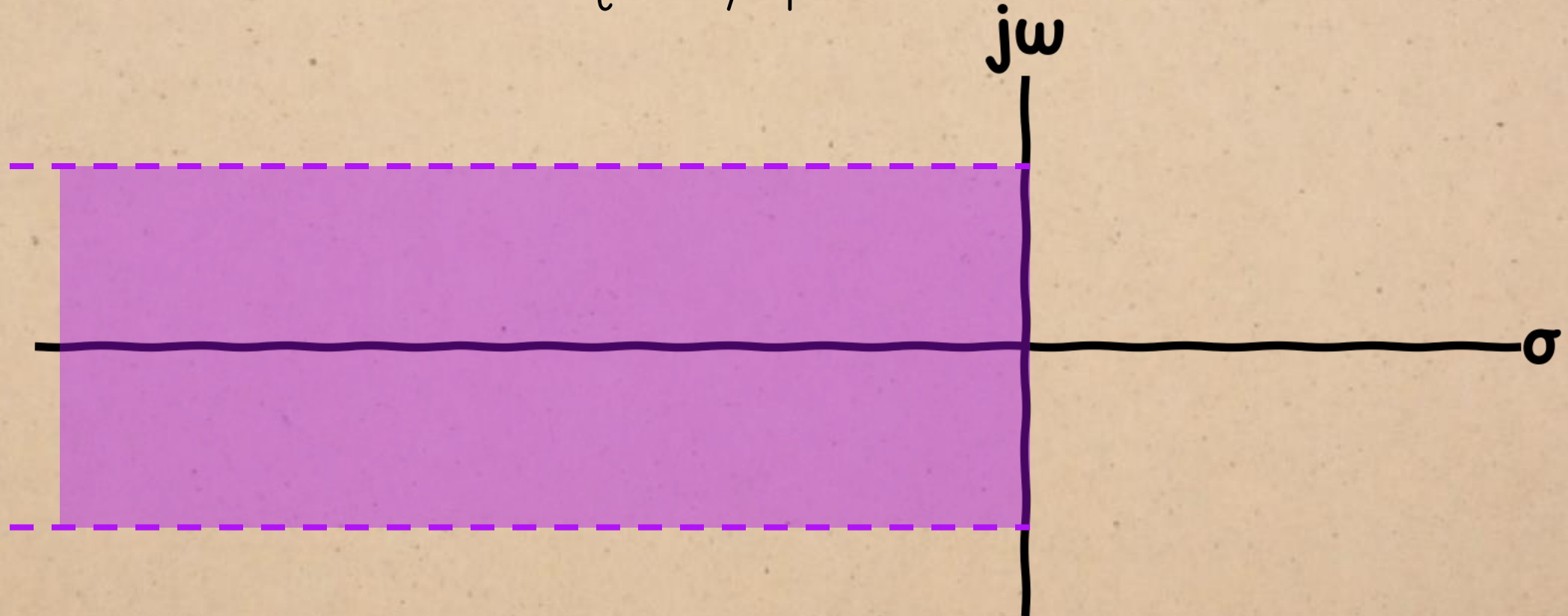
DESIGN GUIDE TO THE S-PLANE

frequency specifications:



DESIGN GUIDE TO THE S-PLANE

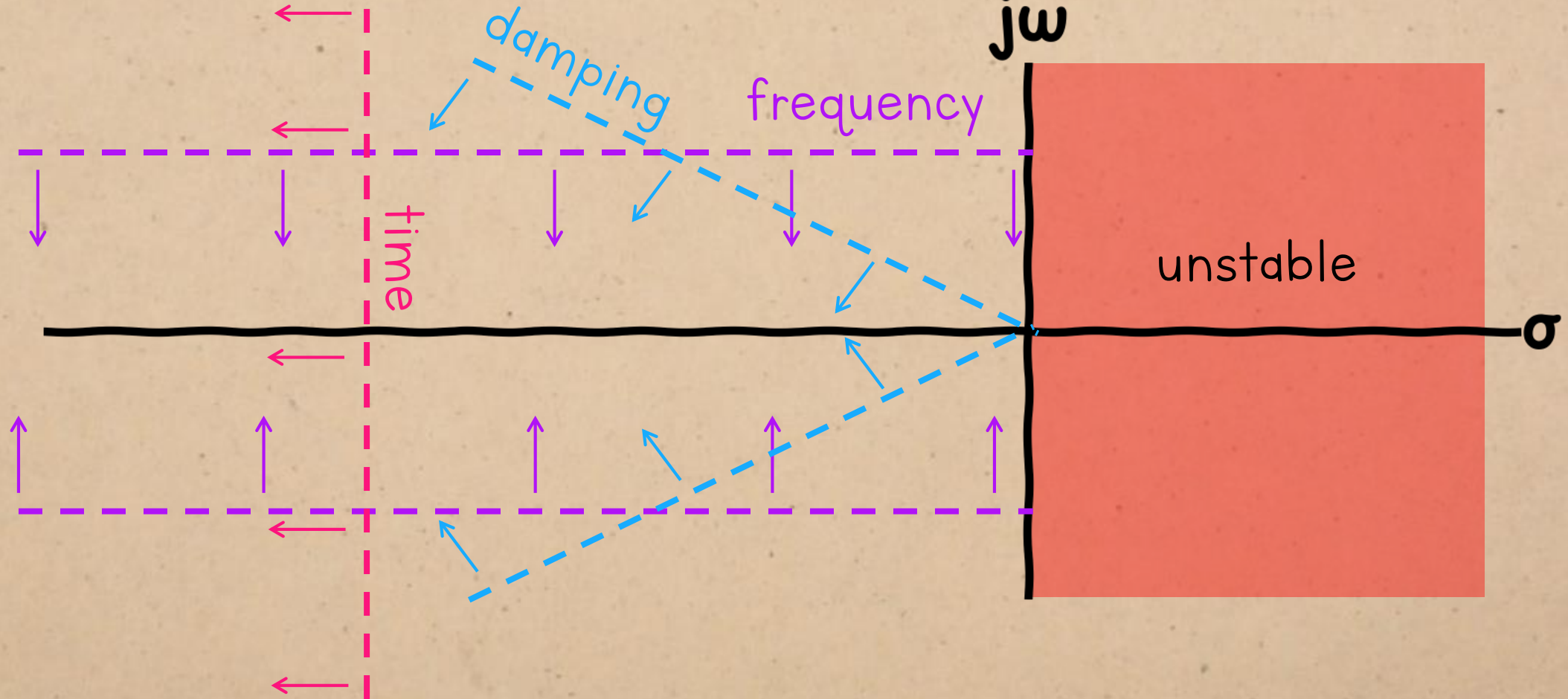
frequency specifications:



A frequency specification restricts poles to a horizontal region.

DESIGN GUIDE TO THE S-PLANE

The complete map:



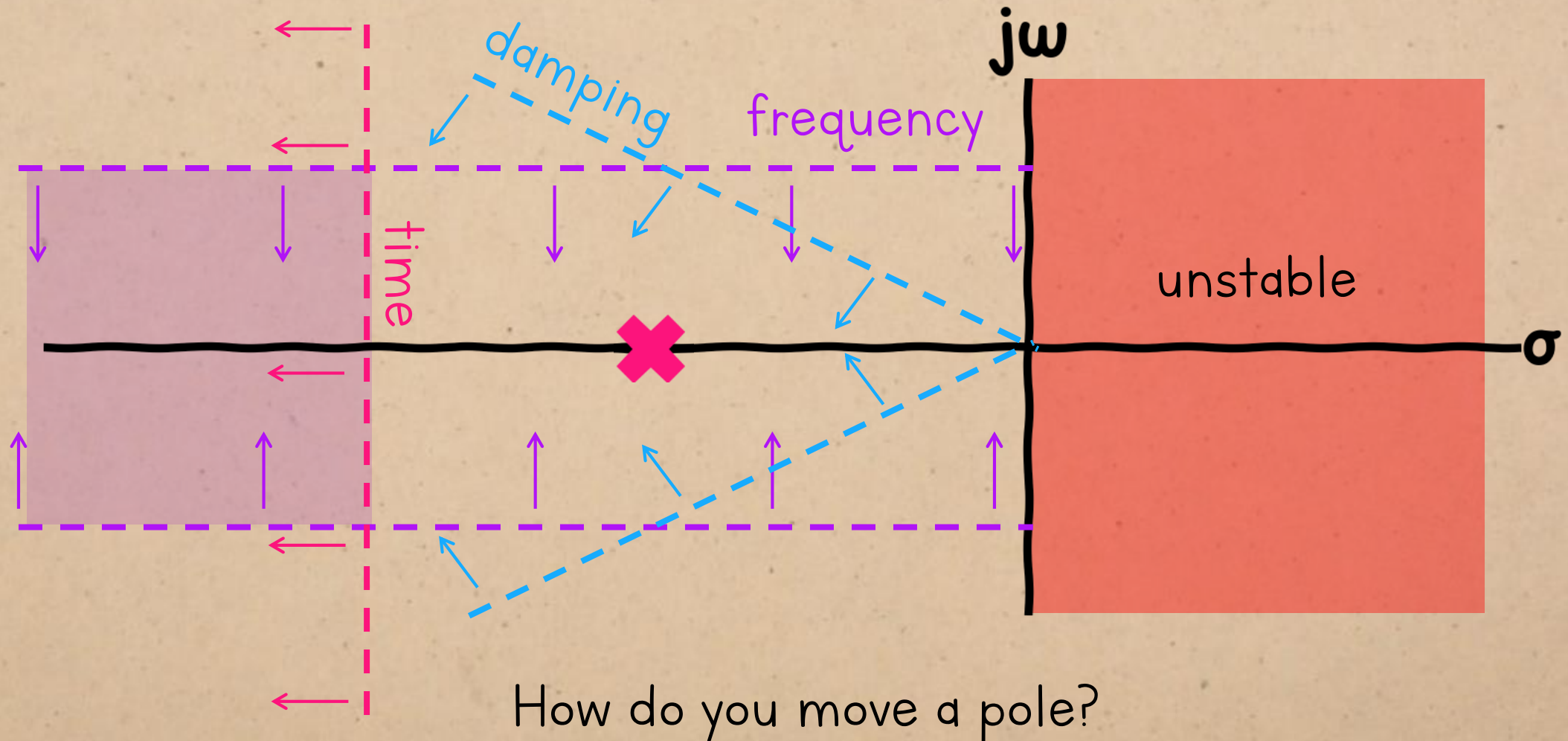
CONTROL SYSTEM ENGINEERING

“Control Engineering is the science and art of
~~causing change to occur in conformity with will.”~~

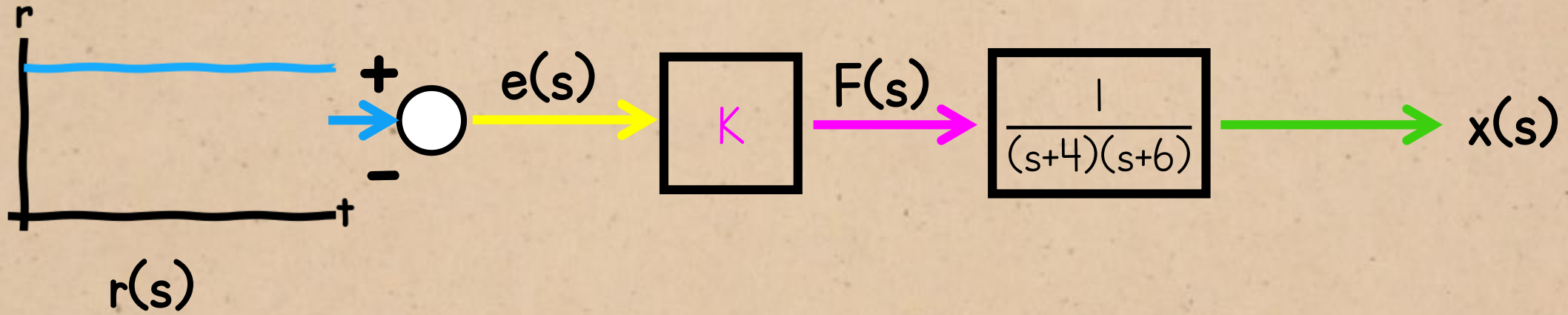
~~choosing inputs to get the outputs that we want.~~

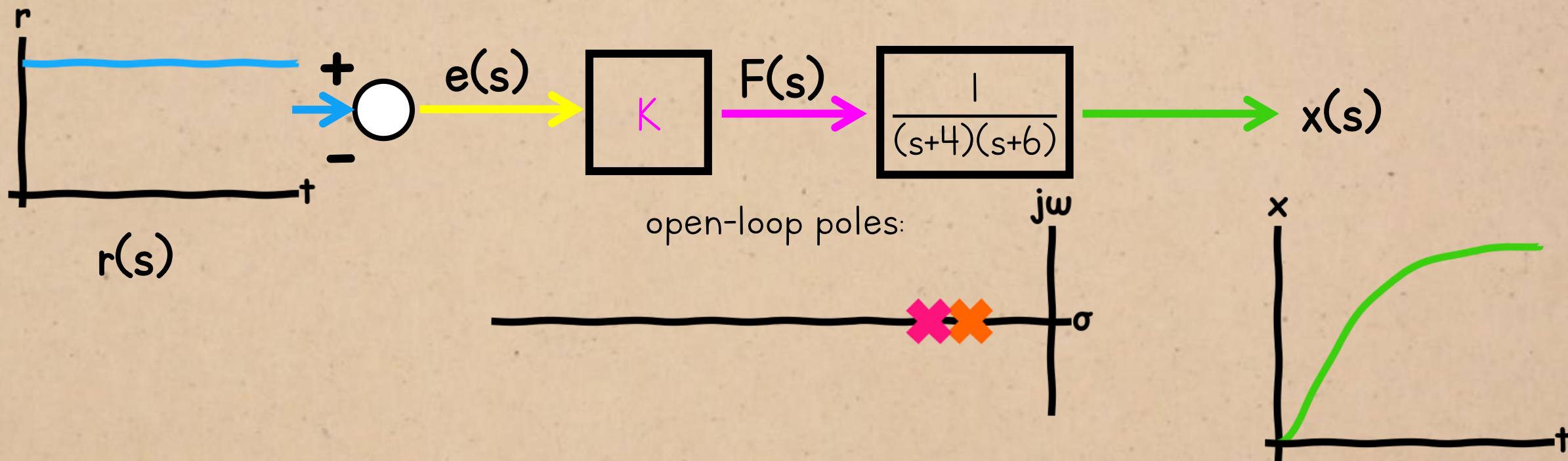
moving the system's poles into the area of
the s-plane that satisfies all specifications.

DESIGN GUIDE TO THE S-PLANE

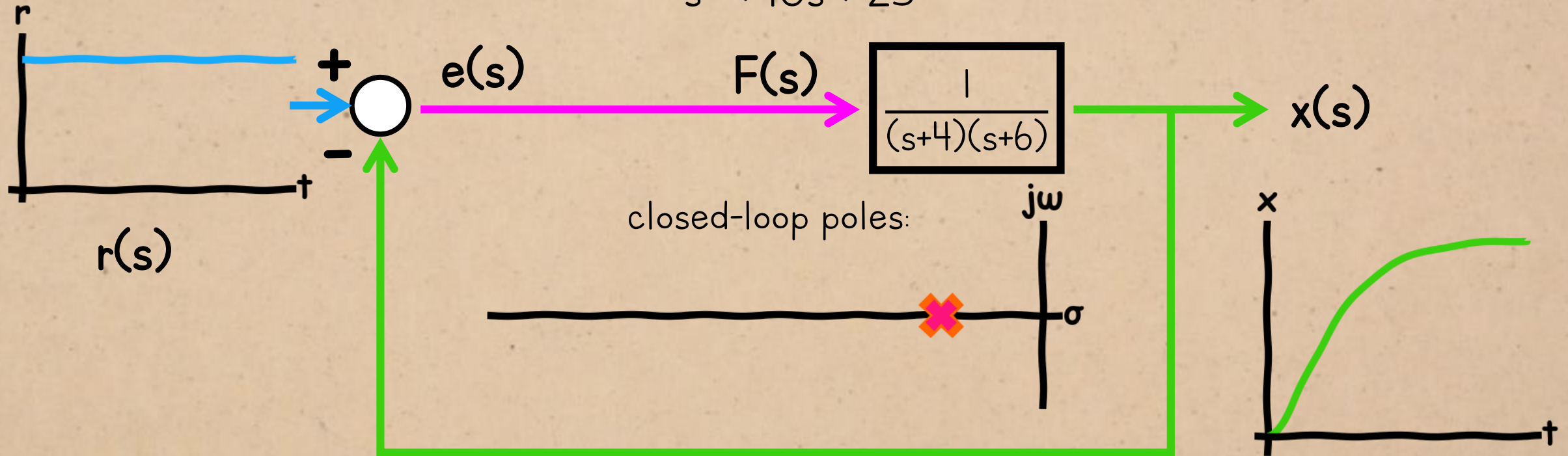


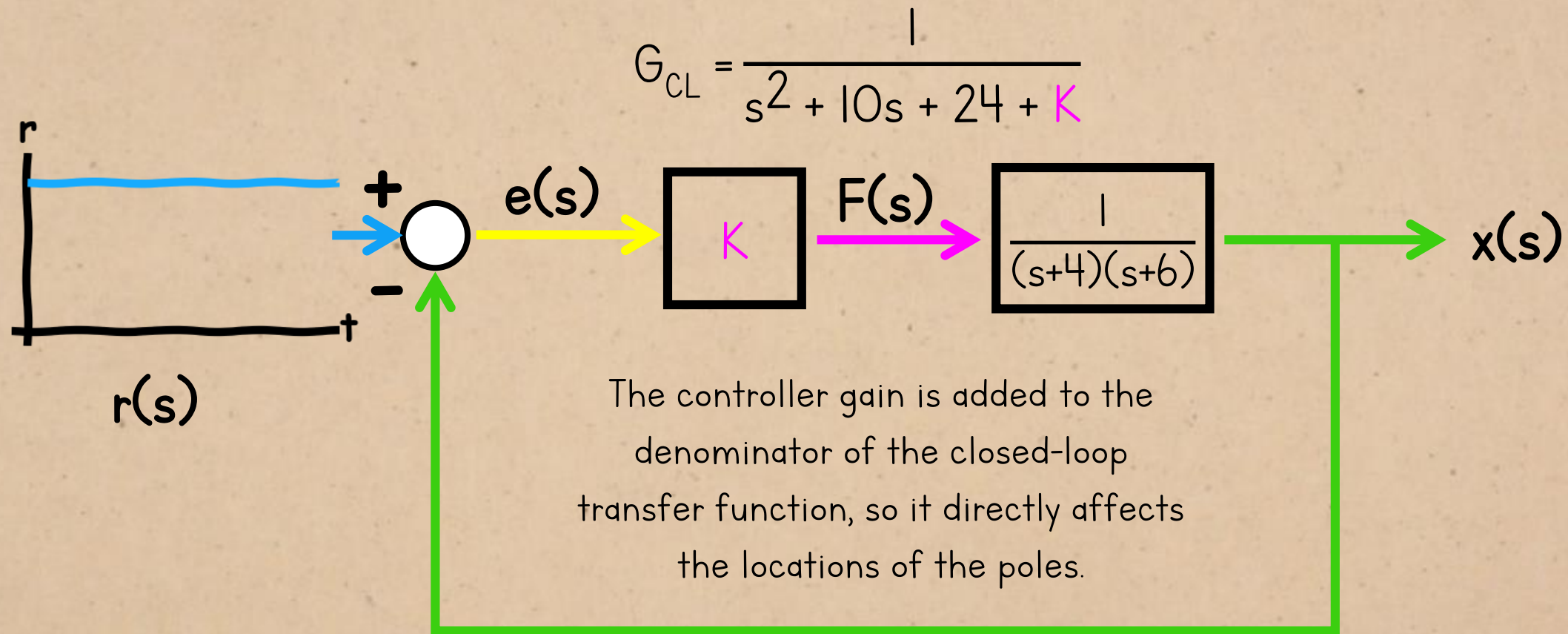
You change the gain:



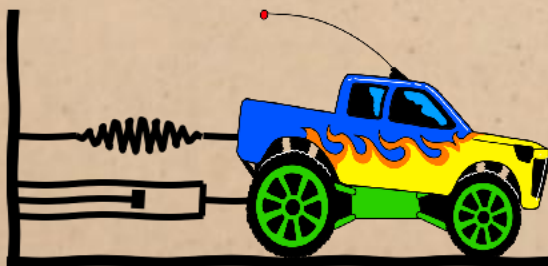
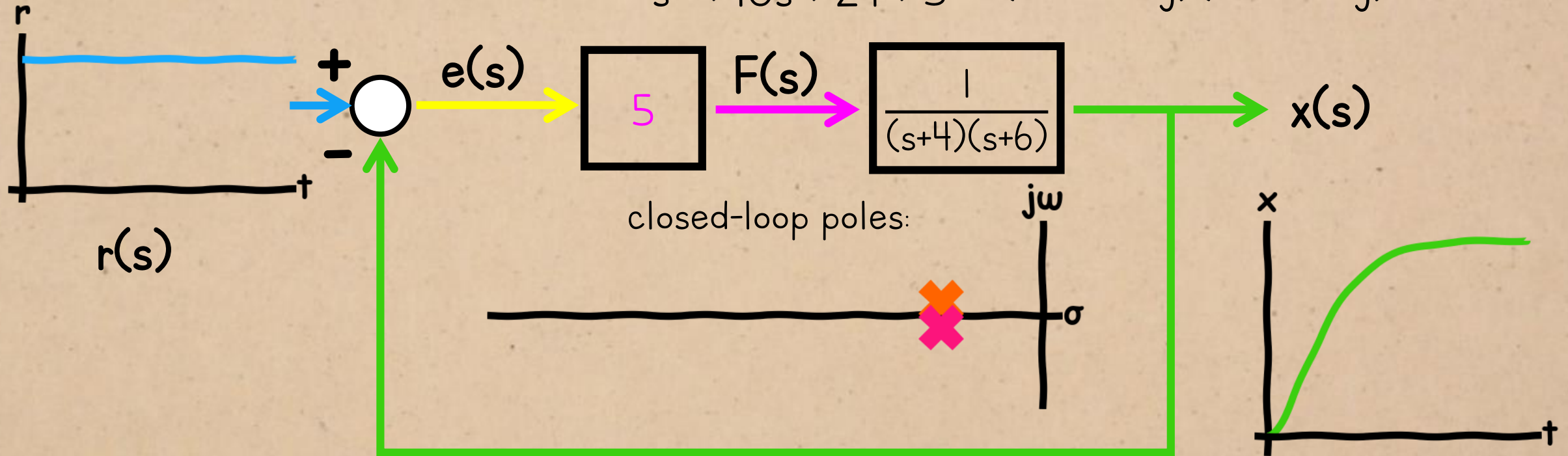


$$G_{CL} = \frac{1}{s^2 + 10s + 25}$$



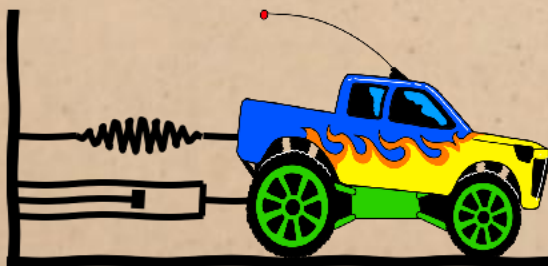
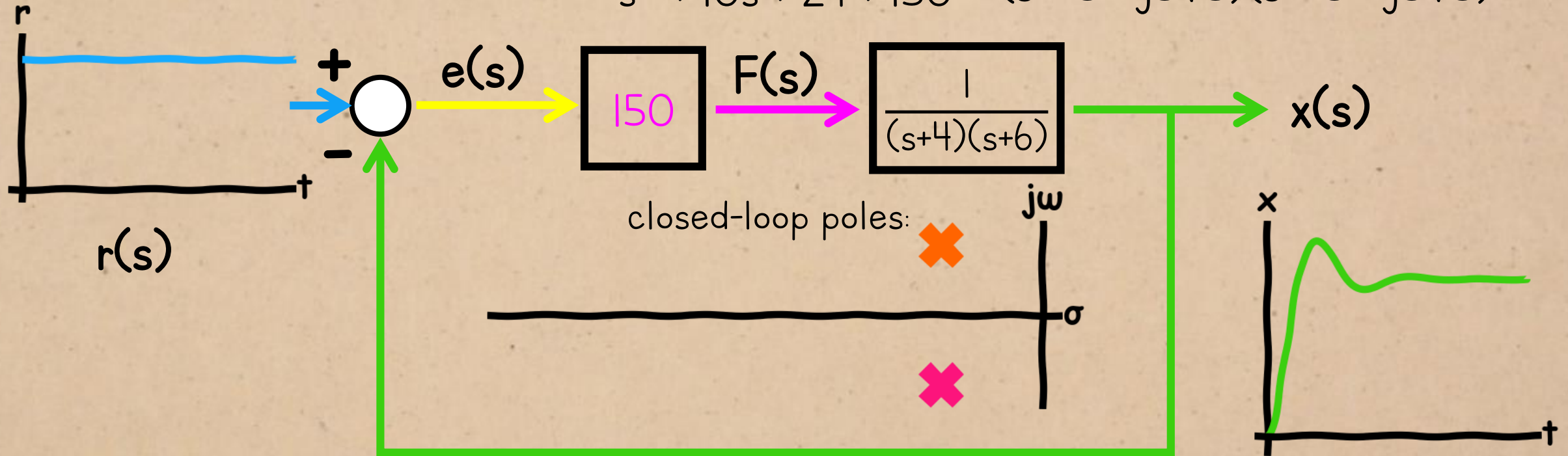


$$G_{CL} = \frac{1}{s^2 + 10s + 24 + 5} = \frac{1}{(s + 5 - 2j)(s + 5 + 2j)}$$

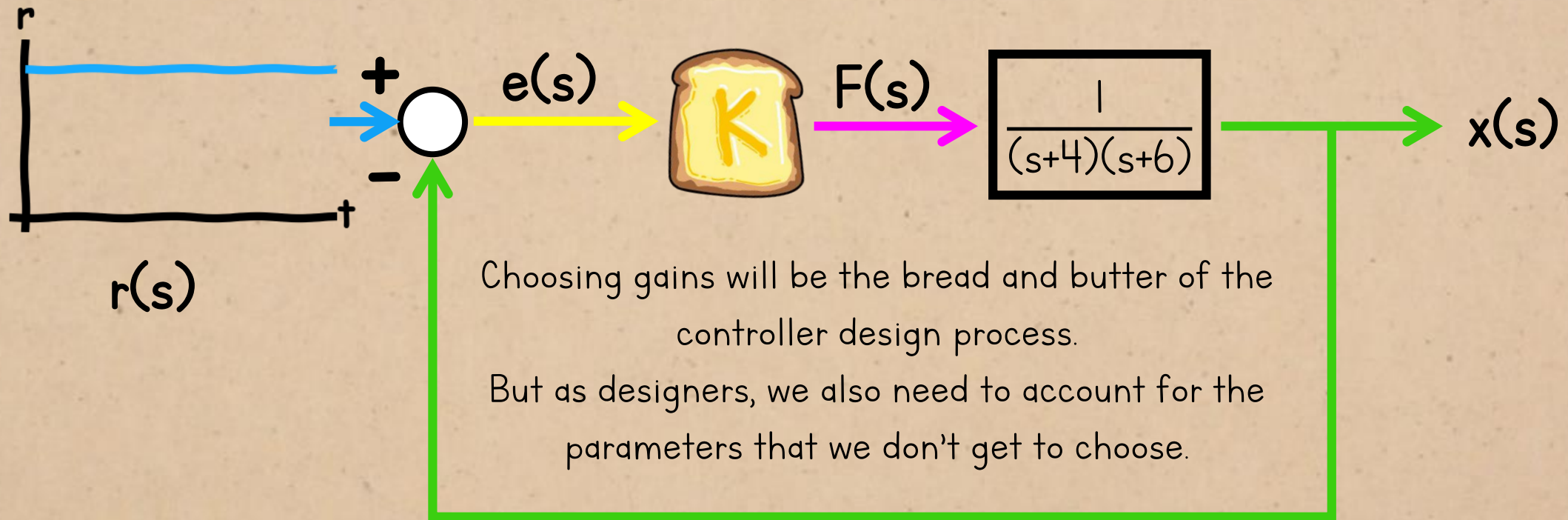


The proportional controller with a gain of five results in complex poles that are close to the real axis, so the response is similar to the critically damped response.

$$G_{CL} = \frac{1}{s^2 + 10s + 24 + 150} = \frac{1}{(s + 5 - j5\sqrt{5})(s + 5 + j5\sqrt{5})}$$



Increasing the gain to 150 gives poles with a larger imaginary part/ smaller damping ratio, so we see a more pronounced oscillation.

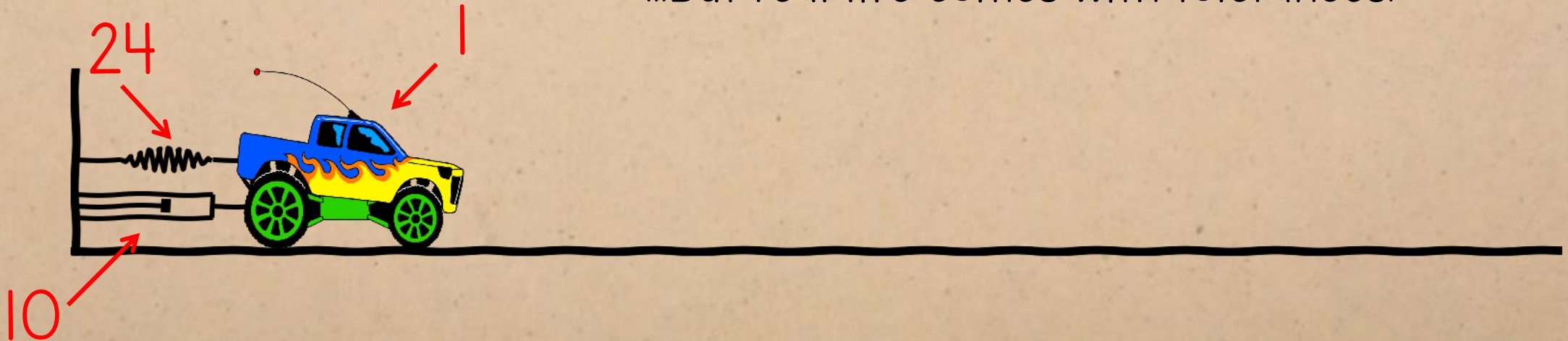


Let $m = 1$, $b = 10$ and $k = 24$:

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

nominal model = model of the plant
where all the parameters are
assigned typical values.

...But real life comes with tolerances.

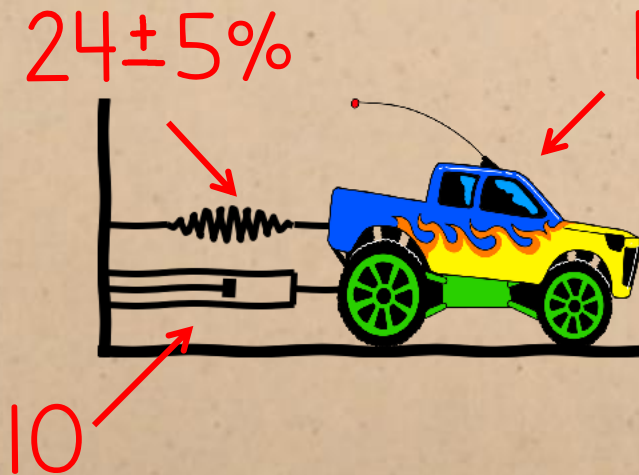


Let $m = 1$, $b = 10$ and $k = 24 \pm 5\%$:

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

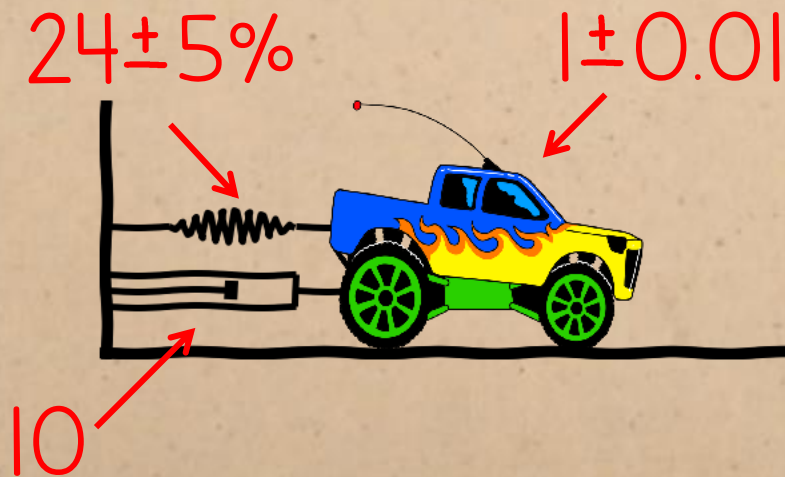
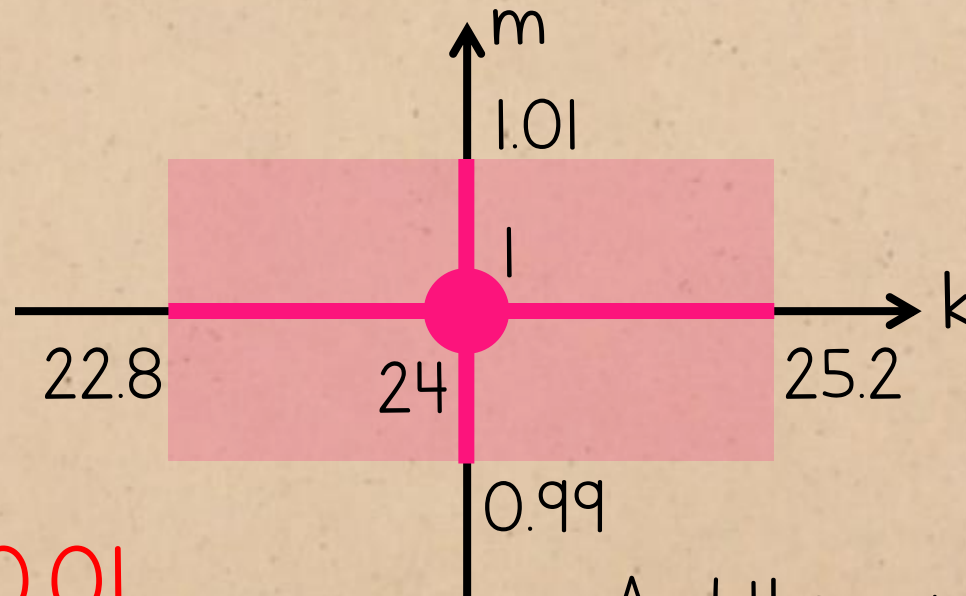


Maybe the spring constant varies by 5% depending on the temperature...



Let $m = 1 \pm 0.01$, $b = 10$ and $k = 24 \pm 5\%$:

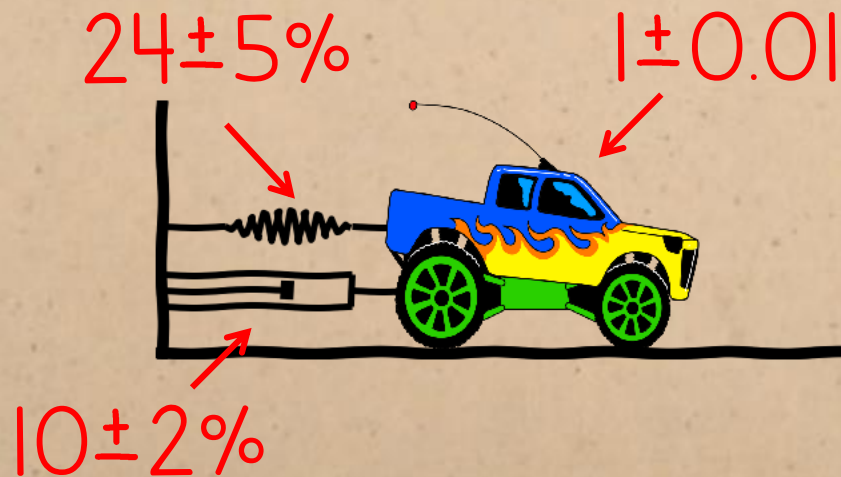
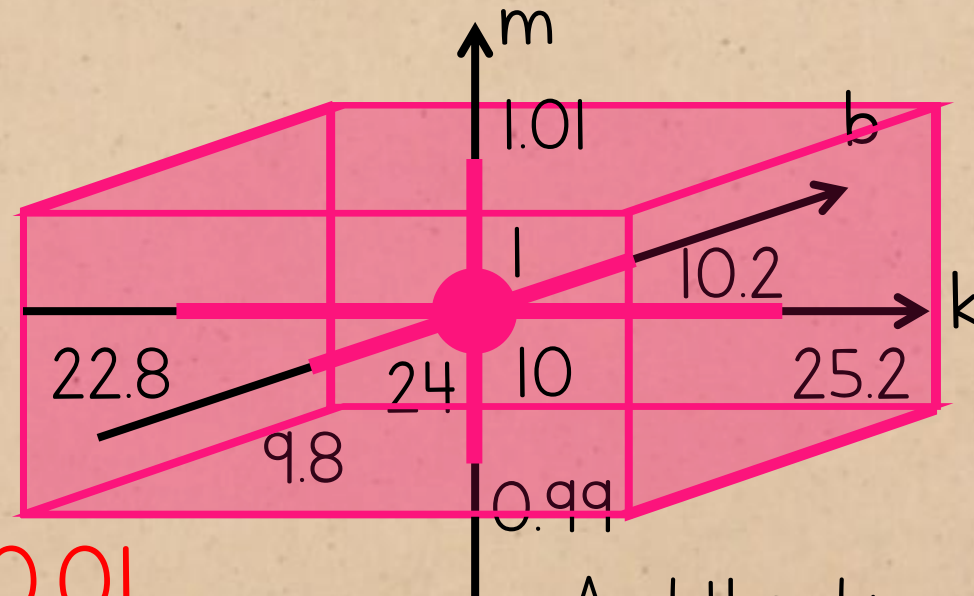
$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$



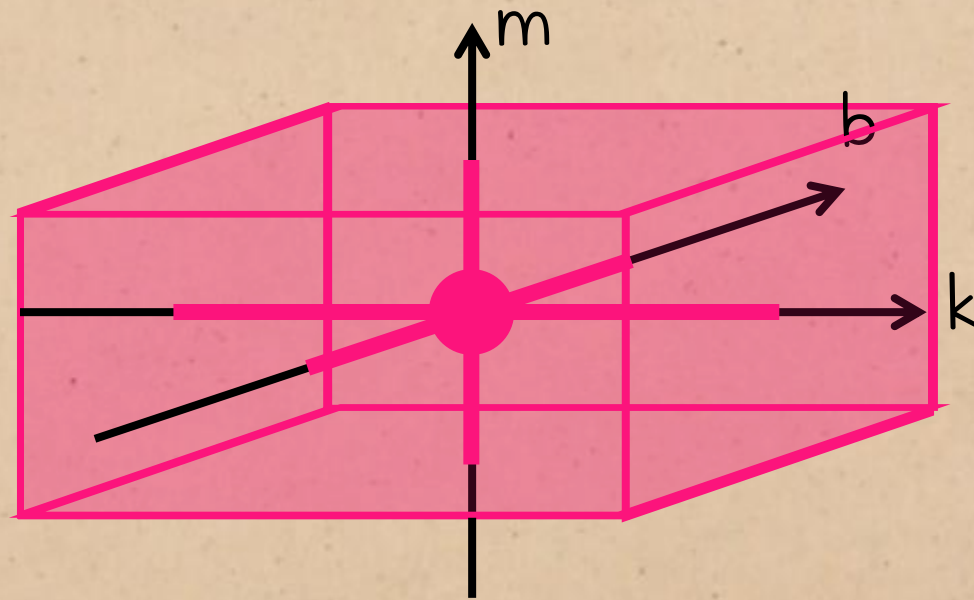
...And the mass might increase or decrease by 10g due to variations in the manufacturing process...

Let $m = 1 \pm 0.01$, $b = 10 \pm 2\%$ and $k = 24 \pm 5\%$:

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

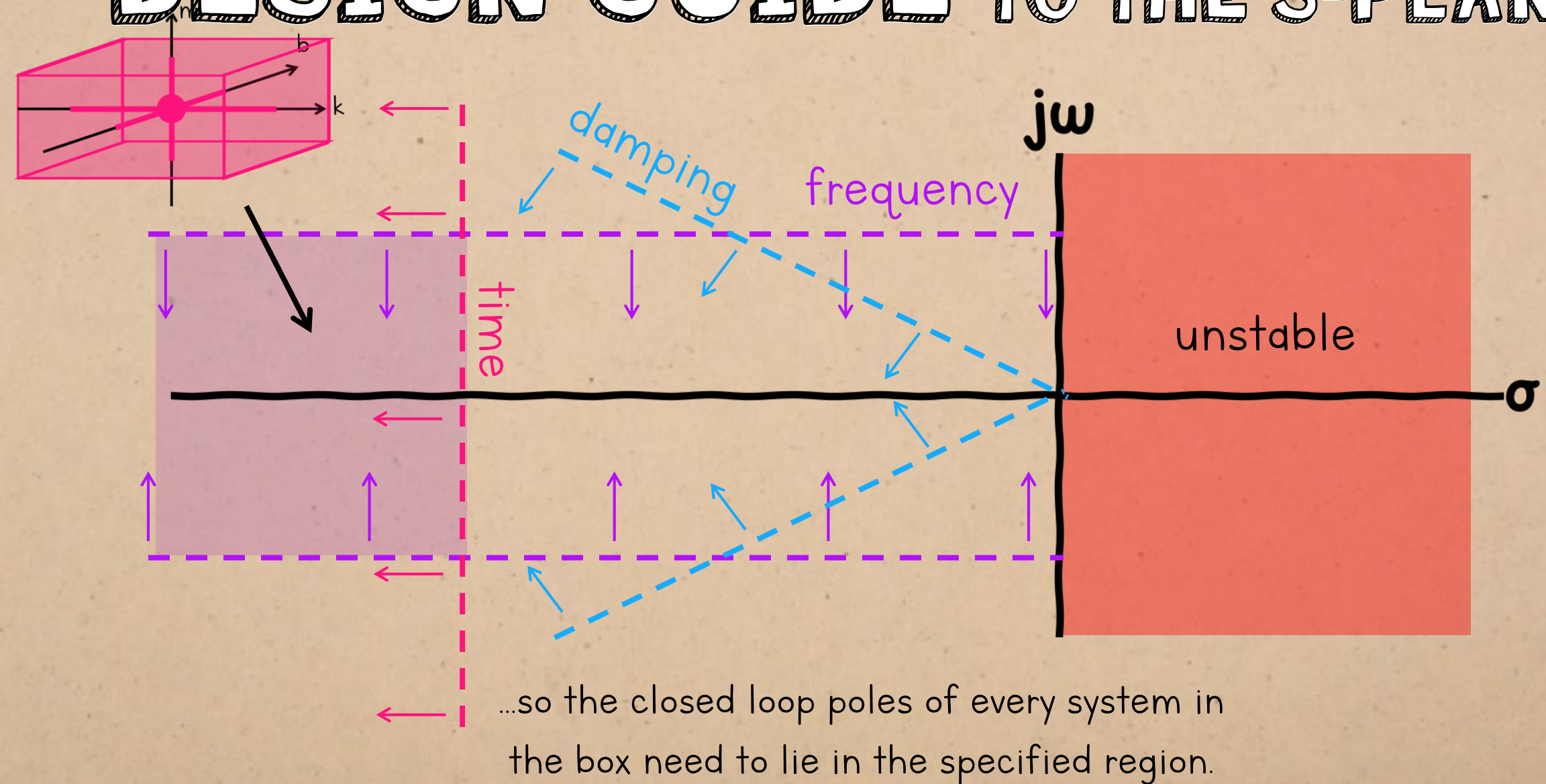


...And the damping coefficient might vary by 2% if the planetary alignment doesn't favour us on a given day.

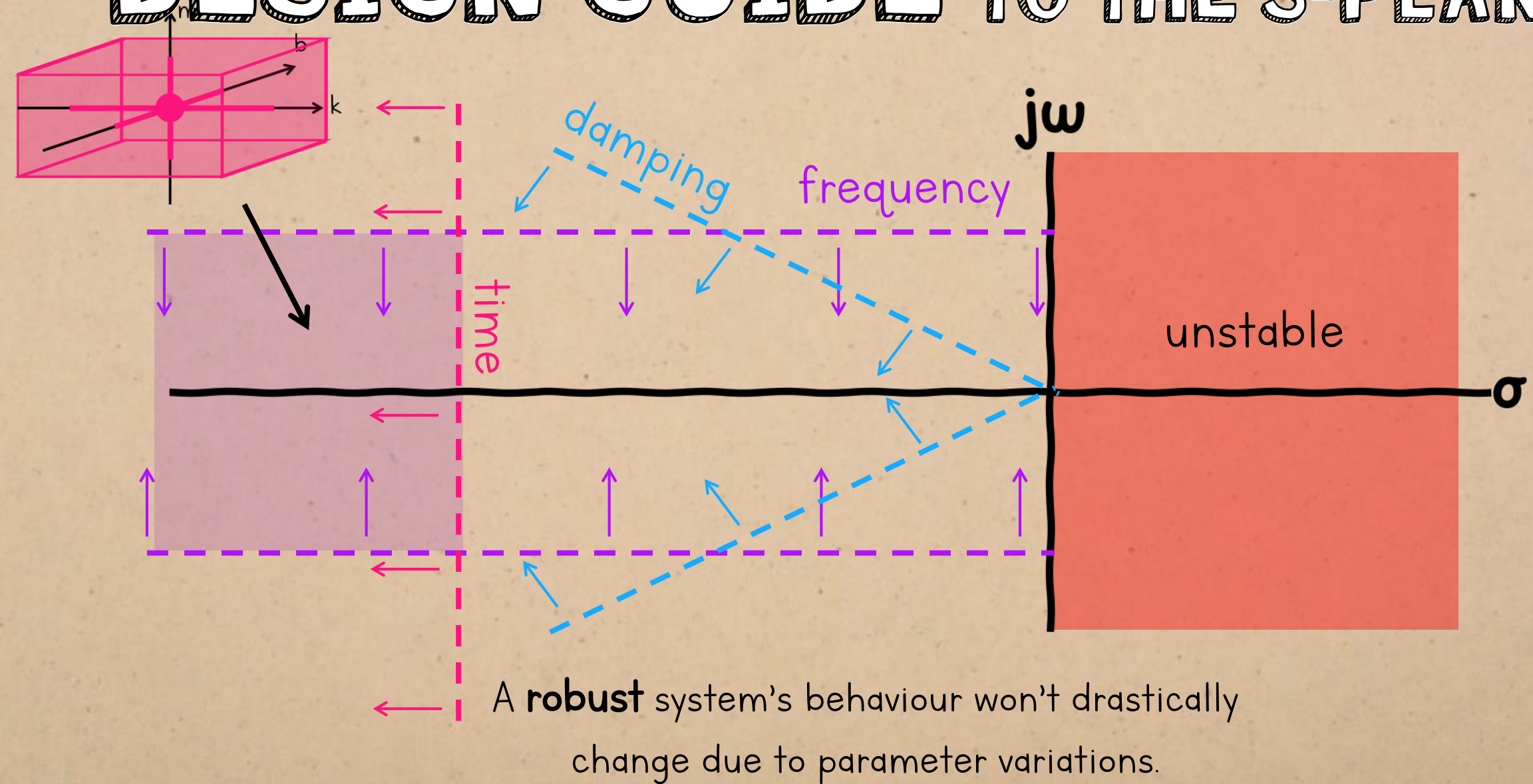


The true plant model could be
anywhere in this box...

DESIGN GUIDE TO THE S-PLANE



DESIGN GUIDE TO THE S-PLANE



SENSITIVITY

= opposite of robustness.

SENSITIVITY

= opposite of robustness.

If a system is **sensitive** to a parameter, small changes in that parameter will result in large changes to the system's response.

SENSITIVITY

Mathematically:

the sensitivity of **a**
with respect to **b**

$$S_b^a = \frac{\frac{\partial a}{\partial b}}{\frac{a}{b}}$$

proportional change in **a**

proportional change in **b**

SENSITIVITY

Example:

Each **class test** = 5% of your class mark

= 40% (at most) of your final mark

SENSITIVITY

Example:

Each **class test** = 5% of your mark

14 + 1/2

So let's say you get your test back,
and decide that the tutor who
marked you didn't do a good job
and owes you an extra half-mark...



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SENSITIVITY

Example:

Each class test = 5% of your mark

$$\frac{14}{20} + \frac{1}{2} = +2.5\%$$



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ING

SENSITIVITY

Example:

Each **class test** = 5% of your mark

$$\begin{aligned} & \frac{14}{20} + \frac{1}{2} \\ & = + 2.5\% \\ & = + 0.125\% \\ & \text{(to class mark)} \end{aligned}$$



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SENSITIVITY

Example:

Each class test = 5% of your mark

$$\begin{aligned} & \frac{14}{20} + \frac{1}{2} \\ & = + 2.5\% \\ & = + 0.125\% \quad (\text{to class mark}) \\ & = + 0.0005\% \quad (\text{to final mark}) \\ & \quad (\text{at most}) \end{aligned}$$



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DEPARTMENT OF ELECTRICAL ENGINEERING

SENSITIVITY

Example:

Each class test = 5% of your mark

$$\begin{aligned} & \frac{14}{20} + \frac{1}{2} \\ & = + 2.5\% \\ & = + 0.125\% \quad (\text{to class mark}) \\ & = + 0.0005\% \quad (\text{to final mark}) \\ & \quad (\text{at most}) \end{aligned}$$



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SENSITIVITY

Example:

Each **class test** = 5% of your class mark

= 40% (at most) of your final mark

the sensitivity of your
final mark with respect
to your **test mark**

$$S_{\text{test}}^{\text{final}} = \frac{0.0005}{2.5}$$
$$= 2 \times 10^{-4}$$

SENSITIVITY

Example:

Each **class test** = 5% of your class mark
= 40% (at most) of your final mark

the sensitivity of your
final mark with respect
to your **test mark**

$$S_{\text{test}}^{\text{final}} = \frac{0.0005}{2.5}$$

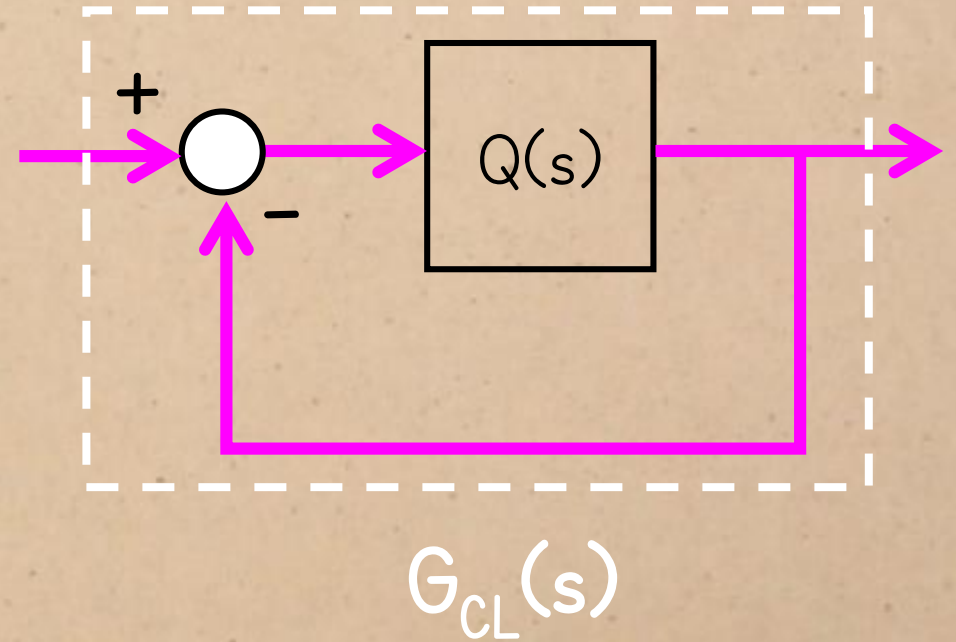
$$= 2 \times 10^{-4}$$

(which probably isn't worth the
effort of sending an email over.)

SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

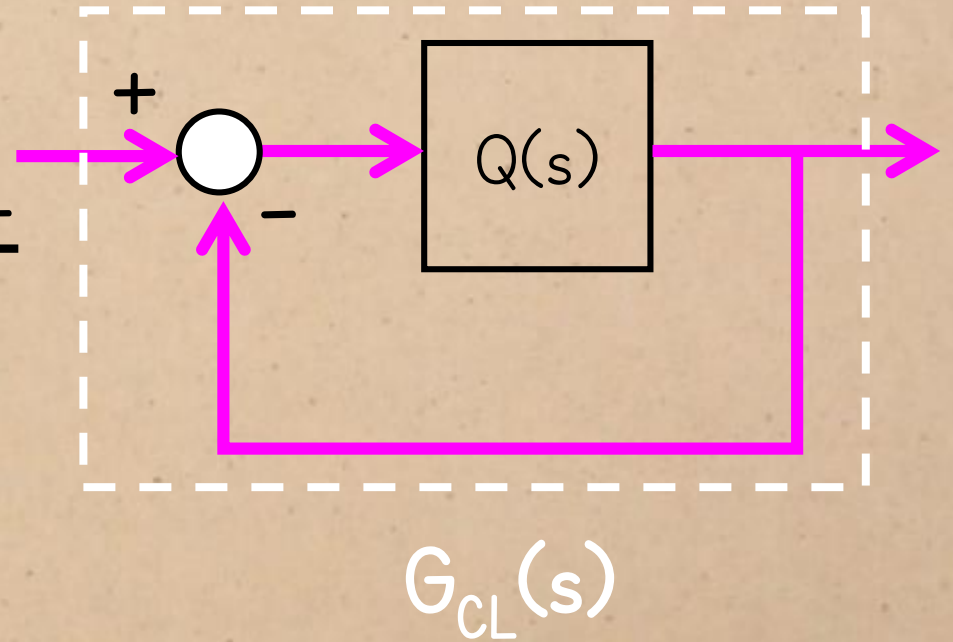
$$S = \frac{\frac{\partial G_{CL}}{G_{CL}}}{\frac{\partial Q}{Q}}$$



SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

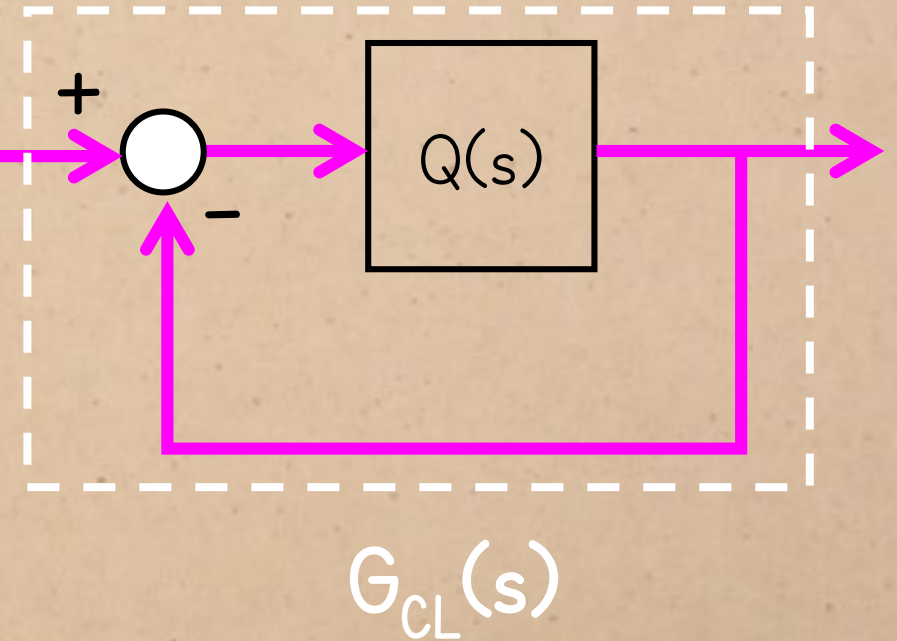
$$S \frac{\partial Q}{Q} = \frac{\partial G_{CL}}{G_{CL}}$$



SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

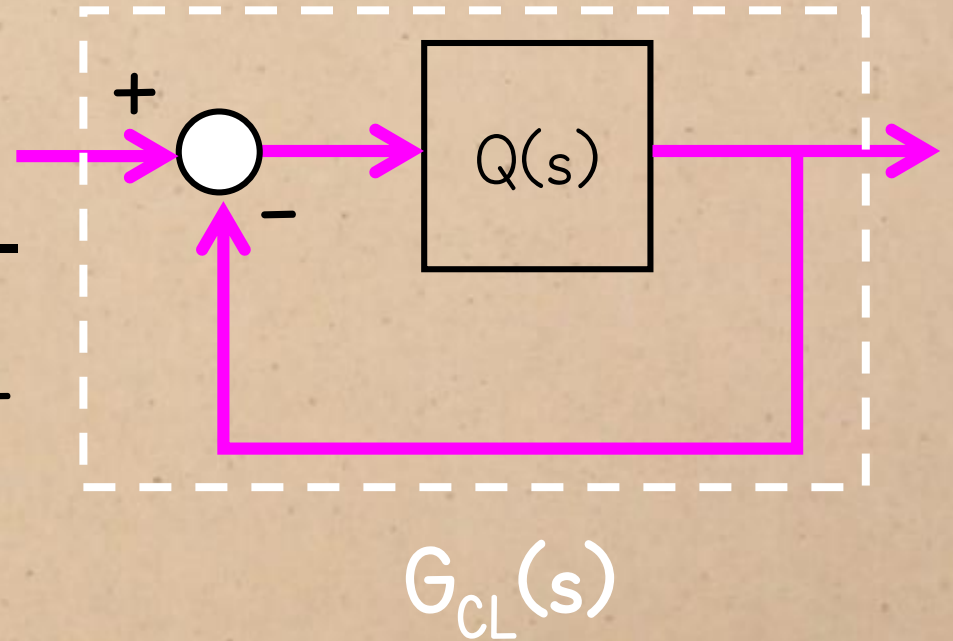
$$S \frac{\partial Q}{\partial G_{CL}} = \frac{Q}{G_{CL}}$$



SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

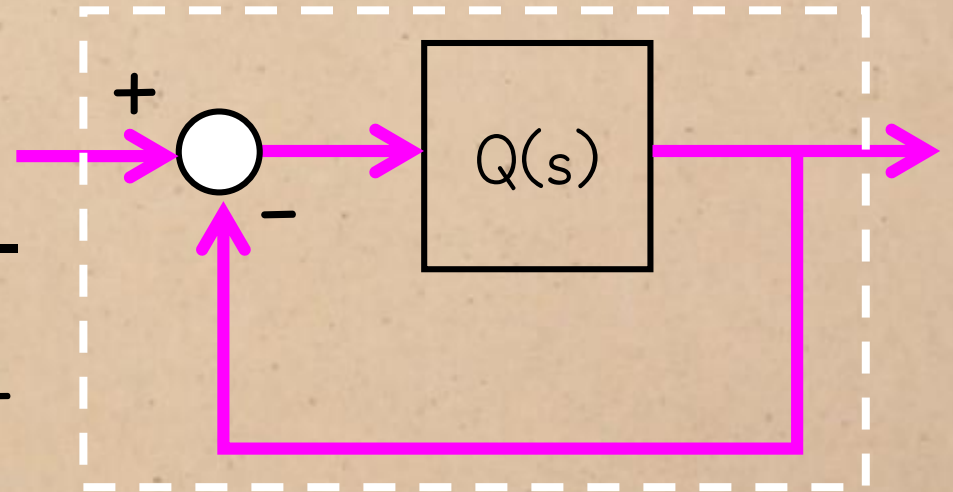
$$S = \frac{\partial G_{CL}}{\partial Q} \frac{Q}{G_{CL}}$$



SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

$$S = \frac{\partial G_{CL}}{\partial Q} \frac{Q}{G_{CL}}$$

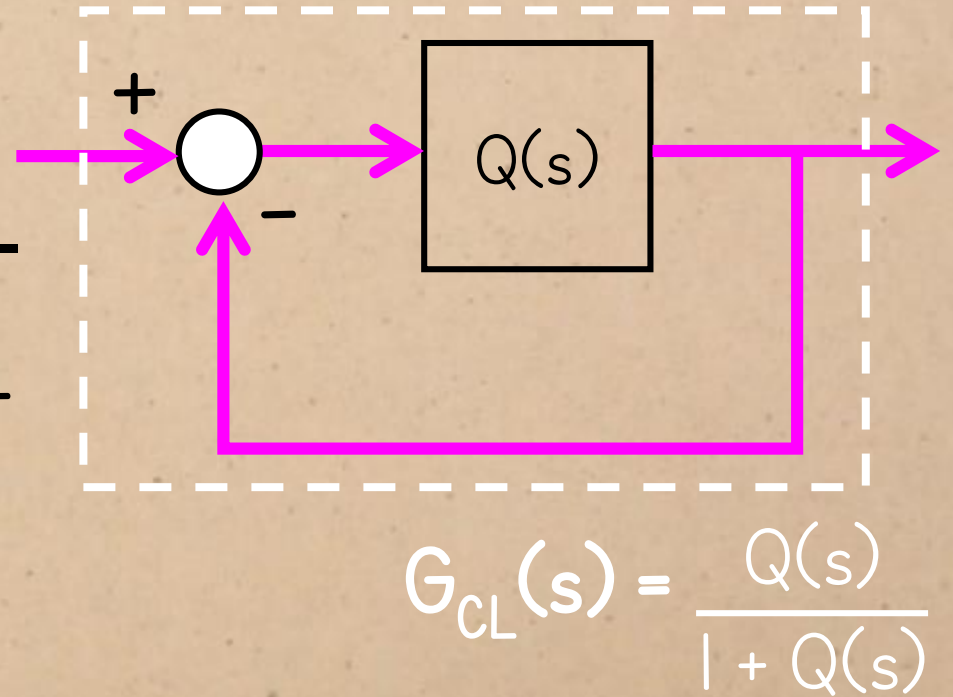


$$G_{CL}(s) = \frac{Q(s)}{1 + Q(s)}$$

SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

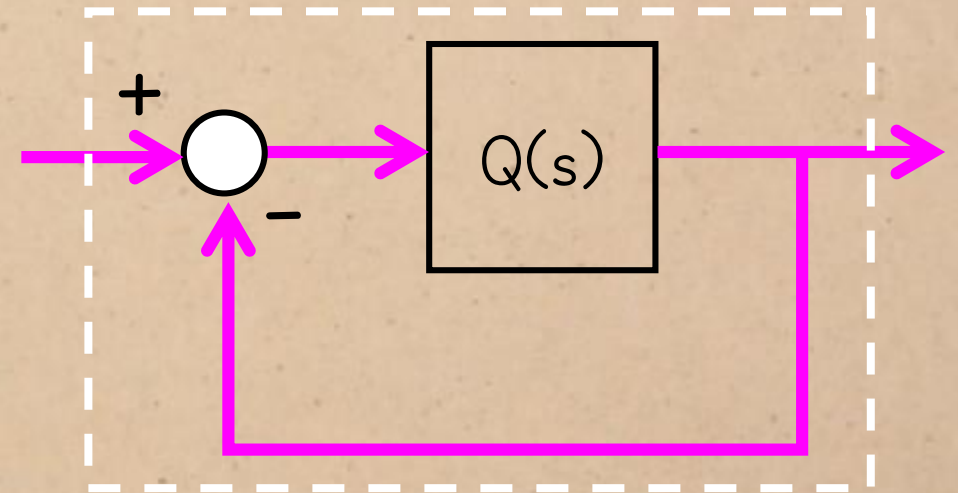
$$S = \frac{\partial G_{CL}}{\partial Q} \frac{Q}{G_{CL}}$$
$$\frac{\partial G_{CL}}{\partial Q} = \frac{1}{(1 + Q(s))^2}$$



SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

$$S = \frac{1}{(1 + Q)^2} \frac{Q}{G_{CL}}$$

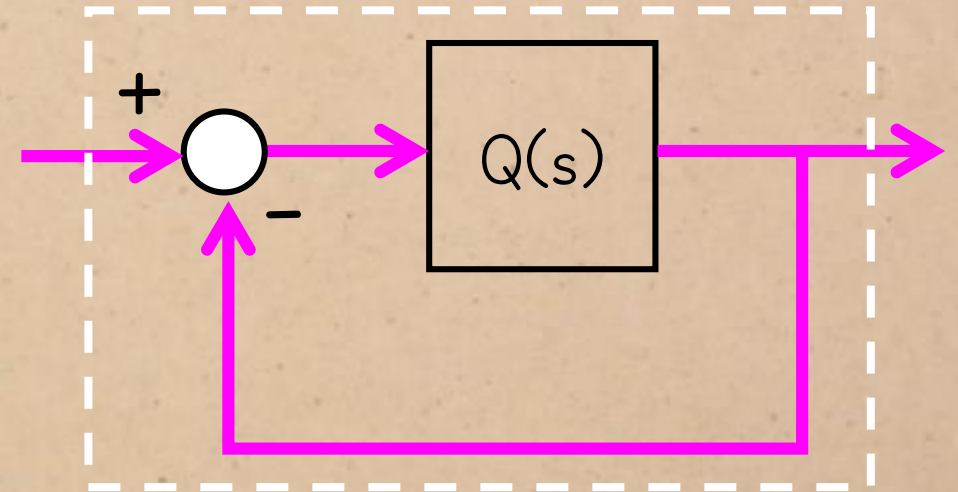


$$G_{CL}(s) = \frac{Q(s)}{1 + Q(s)}$$

SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

$$S = \frac{Q}{(1 + Q)^2} \frac{1}{G_{CL}}$$

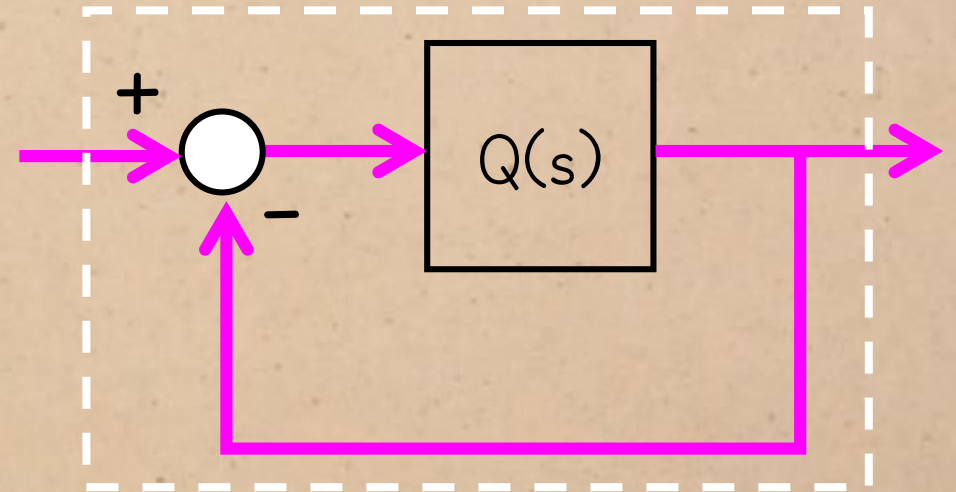


$$G_{CL}(s) = \frac{Q(s)}{1 + Q(s)}$$

SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

$$S = \frac{\cancel{Q}}{(1 + Q(s))^2} \frac{1}{\cancel{\frac{Q}{1+Q}}}$$

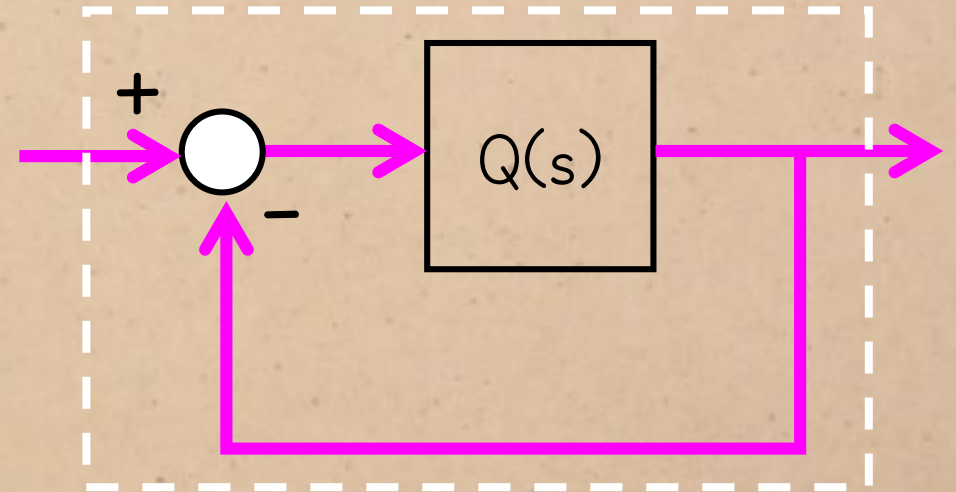


$$G_{CL}(s) = \frac{Q(s)}{1 + Q(s)}$$

SENSITIVITY

The sensitivity of the closed loop transfer function (G_{CL}) with respect to the open loop transfer function (Q):

$$S = \frac{1}{1 + Q(s)}$$



$$G_{CL}(s) = \frac{Q(s)}{1 + Q(s)}$$

SENSITIVITY

sensitivity function:

$$\text{if } S = \frac{1}{1 + Q(s)} > 1$$

the system is **sensitive**

(changes to the open loop will be magnified in closed loop)

SENSITIVITY

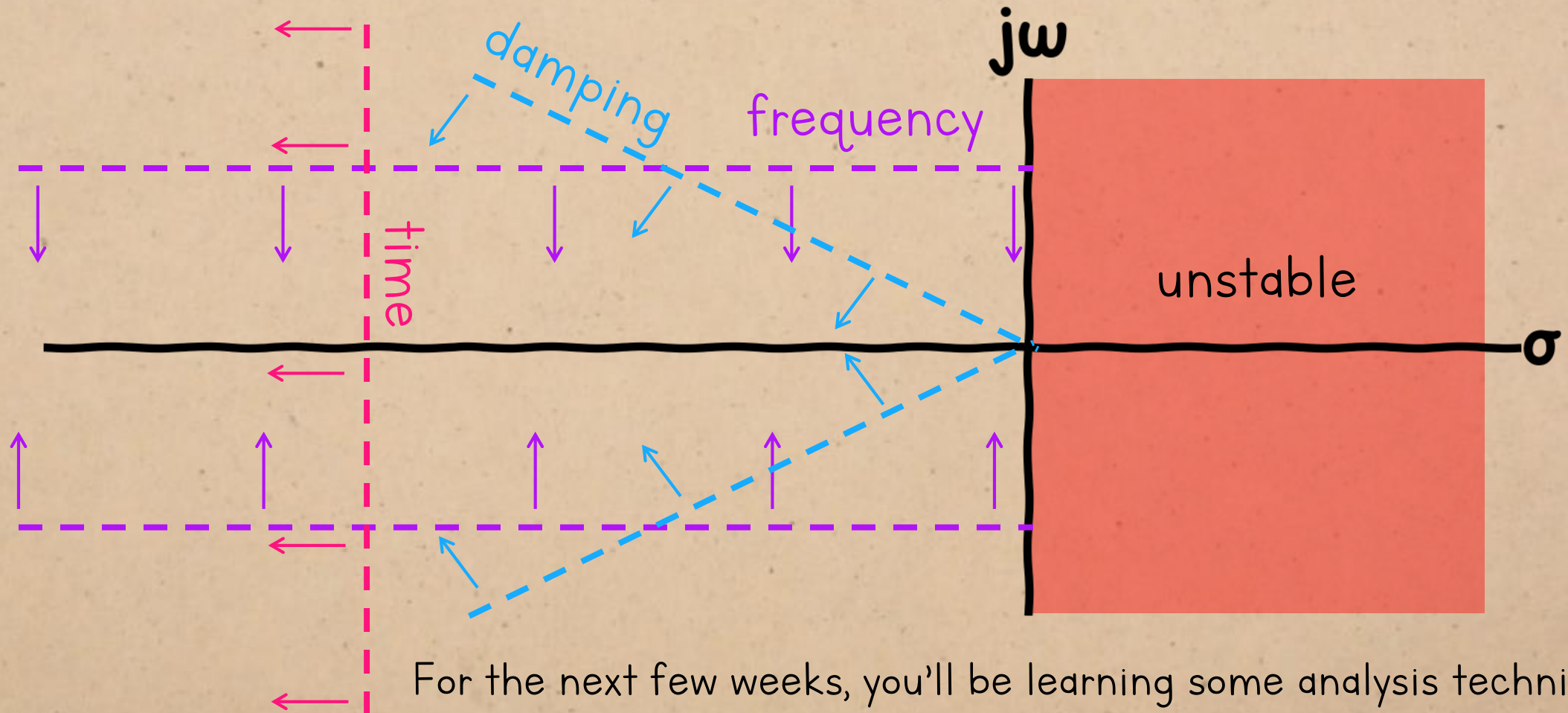
sensitivity function:

$$\text{if } S = \frac{1}{1 + Q(s)} > 1$$

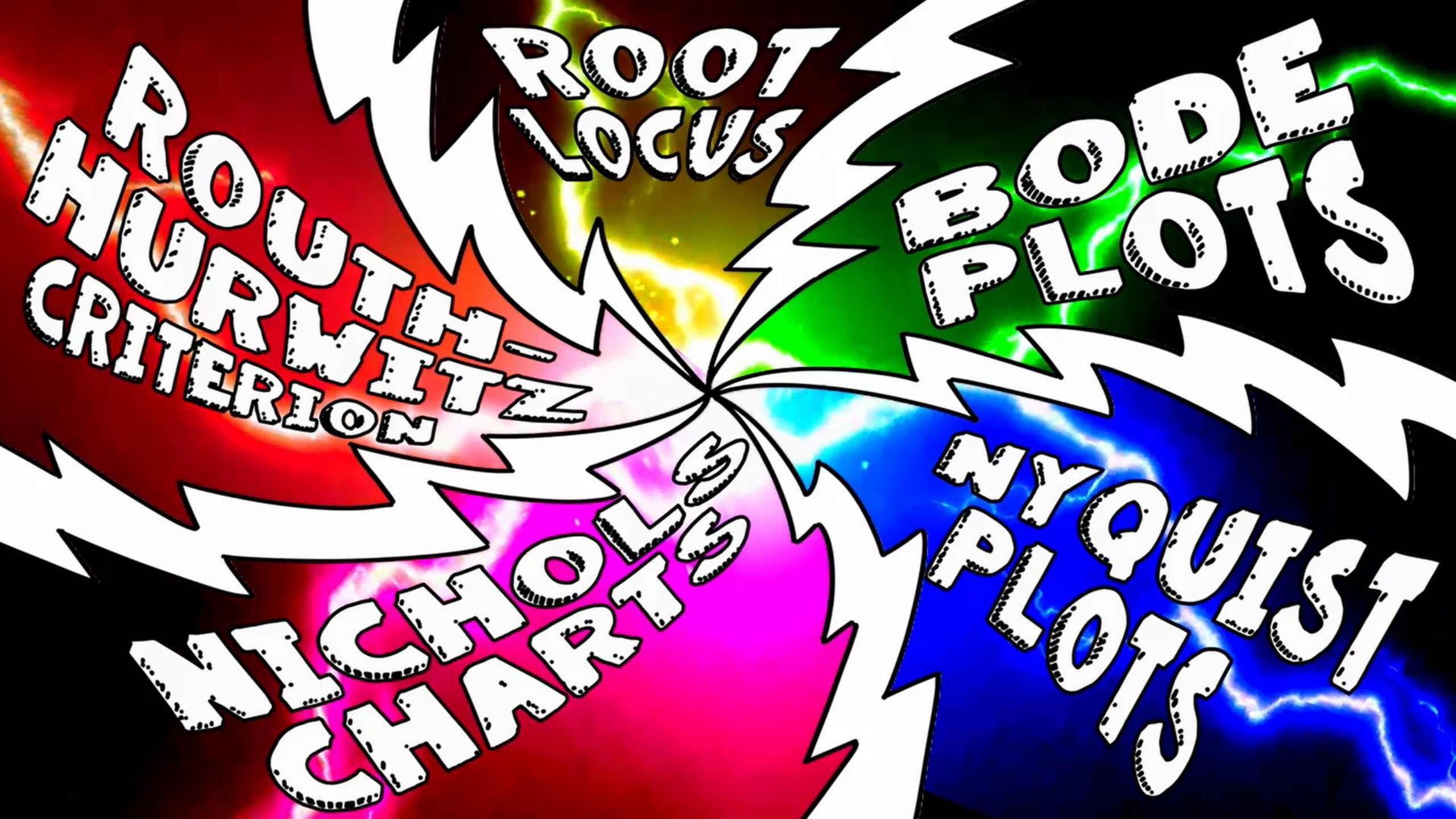
the system is **sensitive**

- ∴ For the system to be **robust**, we must avoid this for all values of s .
(easier said than done, especially at high frequencies)

DESIGN GUIDE TO THE S-PLANE



For the next few weeks, you'll be learning some analysis techniques to help you place the closed-loop poles within this map.



HURWITZ
CRITERION

ROOT
LOCUS

BOODE
PLOTS

NYQUIST
PLOTS

NICHOLS
CHARTS

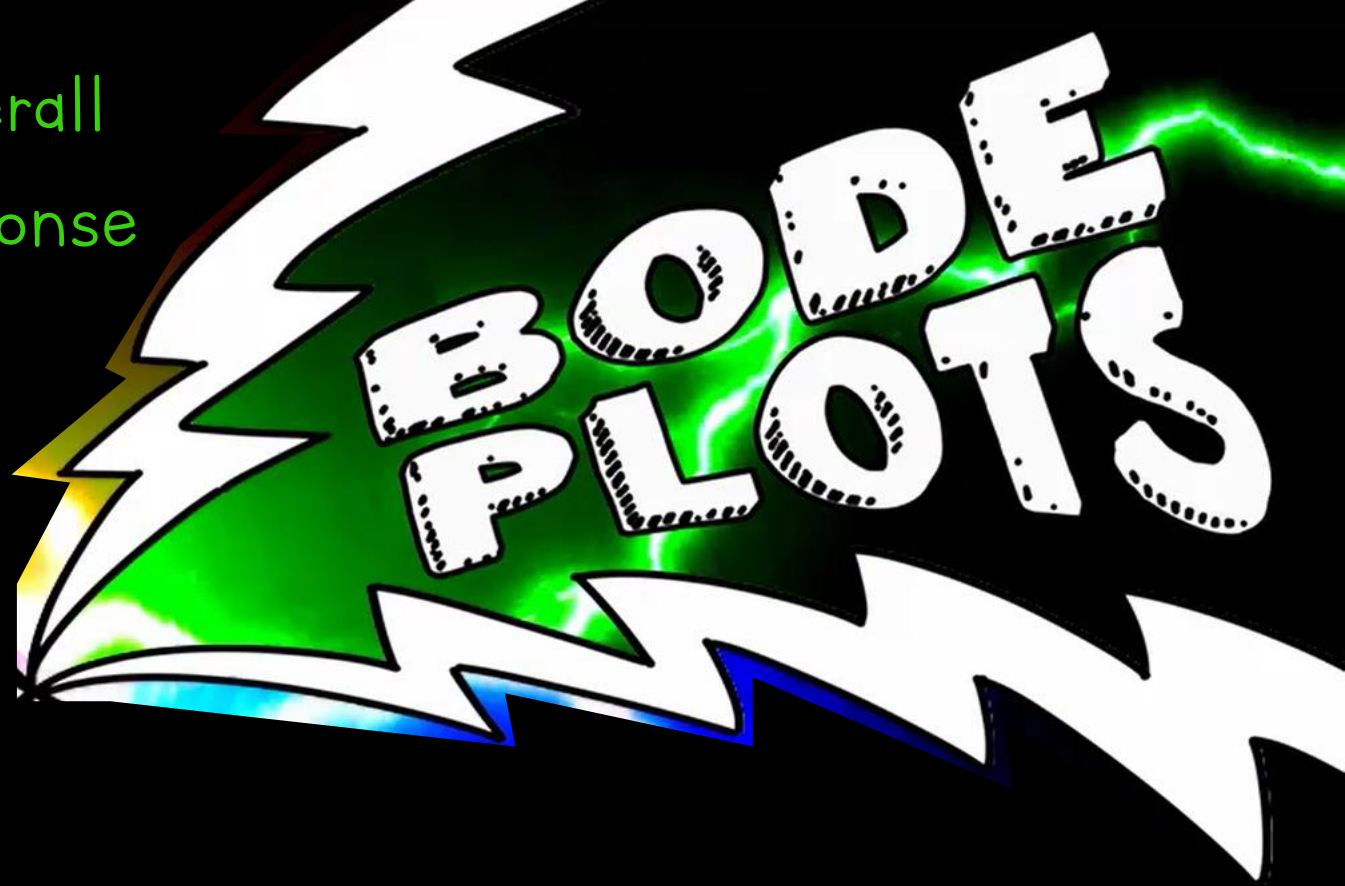


The Routh-Hurwitz criterion, which tells us whether the closed-loop system is stable without having to explicitly calculate the pole locations.

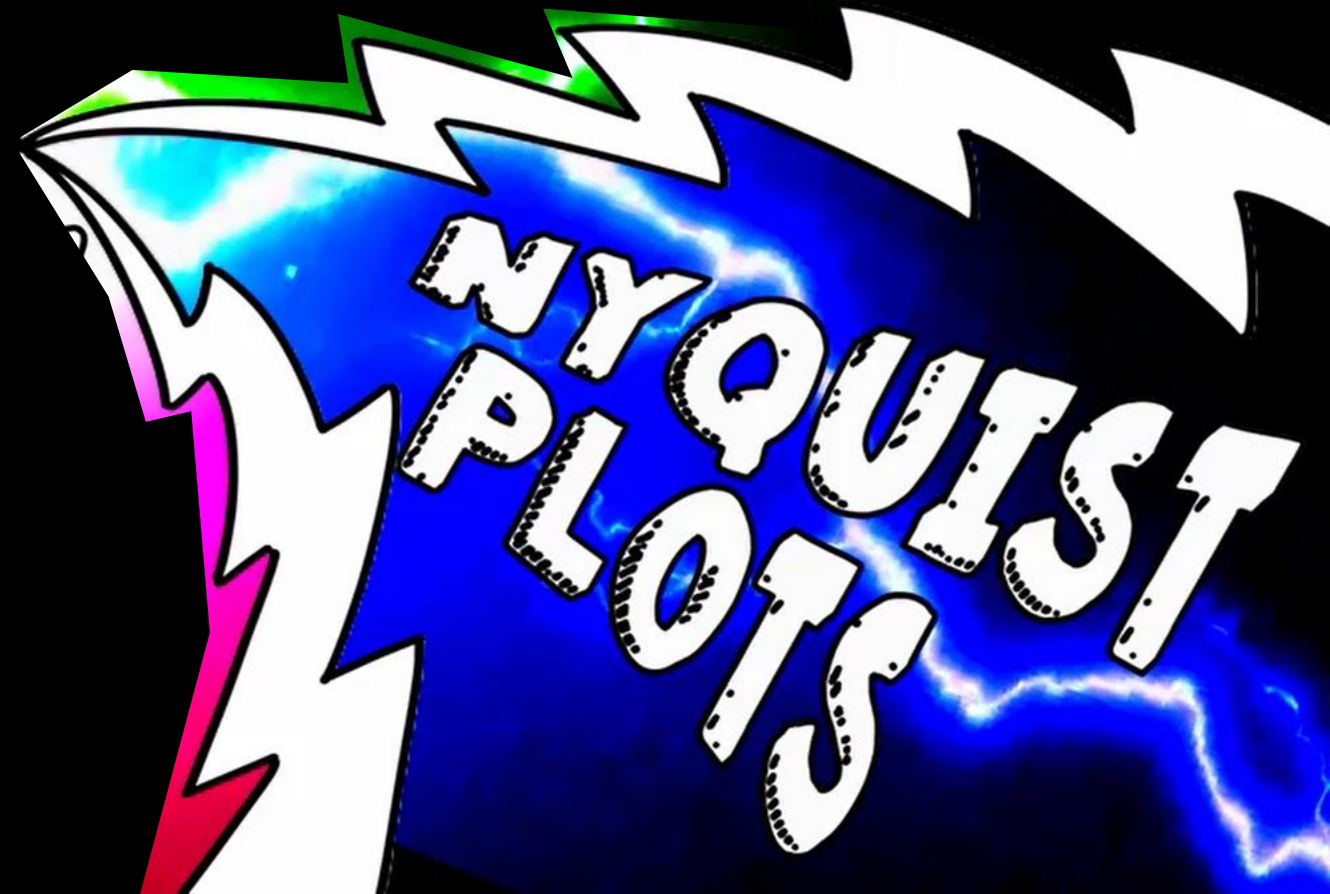


The **root locus**, which tracks how the closed-loop poles move around the s -plane as the gain changes.

Bode plots, which give an overall picture of the frequency response of the system.



Nyquist plots, which indicate how much the system's parameters can change before the system becomes unstable.



...And Nichols charts, which help us evaluate how robust the system is.

